



CURRICULUM

Academic Year 2025– 2026

INFORMATION SCIENCE AND ENGINEERING

III & IV Semester B. E.

Batch-2024

RAMAIAH INSTITUTE OF TECHNOLOGY

(Autonomous Institute, Affiliated to VTU)

(Approved by AICTE, New Delhi & Govt. of Karnataka)

Accredited by NBA & NAAC with A⁺ Grade

About the Institute:

Dr. M. S. Ramaiah a philanthropist, founded 'Gokula Education Foundation' in 1962 with an objective of serving the society. M S Ramaiah Institute of Technology (MSRIT) was established under the aegis of this foundation in the same year, creating a landmark in technical education in India. MSRIT offers 18 UG programs and 15 PG programs. All these programs are approved by AICTE. All eligible UG and PG programs are accredited by National Board of Accreditation (NBA). The institute is accredited with '**A⁺ grade by NAAC in March 2021**' for 5 years. University Grants Commission (UGC) & Visvesvaraya Technological University (VTU) have conferred Autonomous Status to MSRIT for both UG and PG Programs since 2007. The institute has also been conferred autonomous status for Ph.D. program since 2021. The institute is a participant to the Technical Education Quality Improvement Program (TEQIP), an initiative of the Government of India. The institute has 380 competent faculty out of which 70% are doctorates. Some of the distinguished features of MSRIT are: State of the art laboratories, individual computing facility for all faculty members, all research departments active with sponsored funded projects and more than 300 scholars pursuing Ph.D. To promote research culture, the institute has established Centre of Excellence for Imaging Technologies, Centre for Advanced Materials Technology, Centre for Antennas and Radio Frequency systems (CARFS), Center for Cyber Physical Systems, Schneider Centre of Excellence & Centre for Bio and Energy Materials Innovation.

The Entrepreneurship Development Cell (EDC) and "Ramaiah Evolute" have been set up on campus to incubate startups. MSRIT has a strong Placement and Training department with a committed team, a good Mentoring/Proctorial system, a fully equipped Sports department, large air-conditioned library with good collection of book volumes and subscription to International and National Journals. The Digital Library subscribes to online e-journals from Elsevier Science Direct, IEEE, Taylor & Francis, Springer Link, etc. The Institute is a member of DELNET, CMTI and VTU E-Library Consortium. The Institute has a modern auditorium, recording studio, and several hi-tech conference halls with video conferencing facilities. The institute has excellent hostel facilities for boys and girls. MSRIT Alumni have distinguished themselves by occupying high positions in India and abroad and are in touch with the institute through an active Alumni Association.

As per the National Institutional Ranking Framework (NIRF), MoE, Government of India, M S Ramaiah Institute of Technology has achieved 75th rank among 1584 top Engineering Institutions & 31st Rank among 131 Schools of Architecture in India for the year 2025.

About the Department:

The Department of Information Science and Engineering (ISE) was established in the year 1992 with an objective of producing high quality professionals to meet the demands of the emerging field of Information Technology. Department offers Bachelor's program in Information Science and Engineering (B. E), Master's program in Data Science (MTech) and Doctoral program (Ph.D.). The Department of Information Science and Engineering, is a progressive department that has made significant contributions to Academics, Research and Innovation. Under Graduate (UG) is accredited by the National Board of Accreditation in 2001, 2004, 2010, 2015, 2018 and reaccredited in 2022 under Tier-1 till 2028. The department has highly qualified and competent faculty members committed to innovative teaching learning and quality research. Department has 8 well-equipped state of the art laboratories which meets the requirement of curriculum, innovation and research. Collaboration with industries such as Apple, Unisys, Mindtree, Intel, Google, SECO, IBM, NVIDIA etc, has a significant impact on the curriculum, computing infrastructure, teaching & learning and research. The curriculum is centered around Data Science, Artificial Intelligence, IOT, Cloud & Distributed Computing, System Programming, Computer Security and Software development. Curriculum and the teaching learning process ensure that the students demonstrate technical competence, ethical reasoning, creativity in identification & formulation of the problems and develop solutions by using appropriate tools & techniques. Department has established technical clubs/ professional student chapters to provide collaborative learning platform for the students. Echo system has been built to initiate start-ups/Innovation at the department level along with the mentorship program. The activities of the Department led to high profile placements, motivation to become an entrepreneur, and encouragement for higher learning.

VISION OF THE INSTITUTE

To be an Institution of International Eminence, renowned for imparting quality technical education, cutting edge research and innovation to meet global socio-economic needs

MISSION OF THE INSTITUTE

MSRIT shall meet the global socio-economic needs through

1. Imparting quality technical education by nurturing a conducive learning environment through continuous improvement and customization
2. Establishing research clusters in emerging areas in collaboration with globally reputed organizations
3. Establishing innovative skills development, techno-entrepreneurial activities and consultancy for socio-economic needs

QUALITY POLICY

We at MS Ramaiah Institute of Technology strive to deliver comprehensive, continually enhanced, global quality technical and management education through an established Quality Management System complemented by the synergistic interaction of the stake holders concerned

VISION OF THE DEPARTMENT

To evolve as an outstanding education and research center of Information Technology to create high quality Engineering Professionals for the betterment of Society

MISSION OF THE DEPARTMENT

Department of Information Science and Engineering shall create high quality IT Engineering Professionals for the betterment of society by:

1. Providing education through an ever improving curriculum and effective pedagogy techniques.
2. Encouraging extra and co-curricular activities to develop their overall personality along with technical skills.
3. Collaborating with industry and academia for strengthening research, innovation and entrepreneurship ecosystem.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs):

PEO1: Become competent Information Technology professionals with continuous progress in career or learning.

PEO2: Productively engage with society by practicing research or entrepreneurship.

PEO3: Function effectively as professionals in a team environment or individually.

KNOWLEDGE AND ATTITUDE PROFILE (WK)

WK1: A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.

WK2: Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.

WK3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

WK4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.

WK5: Knowledge, including efficient resource use, environmental impacts, whole-life cost, reuse of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.

WK6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

WK7: Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.

WK8: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

WK9: Ethics, inclusive behaviour and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

PROGRAM OUTCOMES (POs):

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WKS)

- PO4: Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
- PO5: Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
- PO6: The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
- PO7: Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
- PO8: Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
- PO9: Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
- PO10: Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
- PO11: Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

PROGRAM SPECIFIC OUTCOMES (PSOs):

- PSO1:** Apply Mathematical models, programming paradigms and software development practices to solve real world problems
- PSO2:** Adopt computing and communication models for developing IT solutions.
- PSO3:** Acquire data engineering skills to develop intelligent systems in a multidisciplinary environment.

**Semester wise Credit Breakdown for B.E Degree Curriculum
Batch 2024-28**

Semester Course Category	First	Second	Third	Fourth	Fifth	Sixth	Seventh	Eighth	Total Credits
Applied Science Course (ASC)	08	08	03	03	--	--	--	--	22
Engineering Science Course (ESC)	09	11	--	--	--	--	--	--	20
Humanities, Social Sciences and Management (HSMC)	02	-	--	--	03	03	--	--	08
Ability Enhancement Course (AEC)	01	01	01	01	01	01	02	--	08
Universal Human Values (UHV)	--	--	02	--	--	--	--	--	02
Professional Core Courses (PCC)	--	--	11	12	12	06	04	--	45
Professional Core Course (IC)	--	--	04	04	03	--	04	--	15
Professional Elective Course (PEC)	--	--	--	--	03	06	03	--	12
Open Elective Course (OEC)	--	--	--	--	--	03	03	--	06
Internship (INT)	--	--	--	00	--	--	--	05	05
Mini Project / Project Work (PW)	--	--	--	--	--	04	--	13	17
Non Credit Mandatory Courses (NCMC)	--	--	Yes	Yes	Yes	--	--	--	--
Total Credits	20	20	21	20	22	23	16	18	160

B.E. in Information Science and Engineering
SCHEME OF TEACHING III SEMESTER Batch:2024

III SEMESTER

Sl. No.	Course Code	Course Name	Teaching Department	Category	Credits				Total contact hours /week			
					L	T	P	Total	L	T	P	S
1.	IS31	Laplace Transforms & Vector Space	Maths	ASC	2	1	0	3	2	2	0	2
2.	IS32	Digital Design and Computer Organization	ISE	PCC(IC)	3	0	1	4	3	0	2	3
3.	IS33	Data Structures	ISE	PCC	3	0	0	3	3	0	0	3
4.	IS34	Operating Systems	ISE	PCC	3	0	0	3	3	0	0	3
5.	IS35	Discrete Mathematical Structures	ISE	PCC	2	1	0	3	2	2	0	2
6.	ISL36	Data Structures Lab	ISE	PCC	0	0	1	1	0	0	2	0
7.	ISL37	Unix System Programming Lab	ISE	PCC	0	0	1	1	0	0	2	0
8.	UHV38	Universal Human Values	ISE	UHV	2	0	0	2	2	0	0	2
9.	ISAEC39X	Ability Enhancement Course- III	ISE	AEC	1	0	0	1	1	0	0	1
Total					16	2	3	21	16	4	6	16
10.	PE69	Physical Education		NCMC	All students have to register compulsorily for any one of the courses with the concerned coordinator (Yoga Teacher/ Physical Education Director/ NSS Coordinator) in the beginning of the III semester. Attending the registered course from III to VI semesters. Successful completion of the registered course and requisite CIE score is mandatory for the award of the degree.							
	YO69	Yoga										
	NS69	NSS										
11.	AM31	Additional Mathematics - I *	Maths	NCMC	0	0	0	0	3			

Ability Enhancement Course- III

S.No.	Course Code	Course Name
1.	ISAEC391	Green IT and sustainability
2.	ISAEC392	Capacity Planning for IT
3.	ISAEC393	Numerical analysis with python

Nomenclature: **ASC:** Applied Science Course, **IC:** Integrated Course (Practical Integrated with Theory Course), **PCC:** Professional Core Course, **UHV:** Universal Human Value Course, **AEC:** Ability Enhancement Course, **NCMC:** Non-credit Mandatory Course

Credit Definition:

- **1 credit = 1 hour/week of lecture (L)**
- **1 credit = 2 hours/week of tutorial (T)**
- **1 credit = 2 hours/week of practical(P)**
- **1 credit = 1 hour/week of Self Learning Hours (S)**

The Program Core Course(IC) is an Integrated course, combining theory with practical components.

- Theory sessions shall be conducted for 3 hours per week, while the practical sessions shall be conducted for 2 hours per week.
- Theory components shall be evaluated through both Continuous Internal Evaluation (CIE) and Semester End Examination (SEE).
- The practical component shall be assessed only through CIE (No SEE).
- However, questions from the practical component of the Integrated course (IC) can be included in the SEE question paper.

The Non Credit Mandatory Course, Physical Education (Sport and Athletics)/Yoga/National Service Scheme (NSS):

1. Student shall select any one of the NCMC's namely, Physical Education (Sport and Athletics)/Yoga/ NSS prescribed for four semesters and shall attend the course from the III semester to VI semester to complete all the formalities of the course and earn requisite CIE score. The Pass/Fail shall be reflected in the VI semester grade card.
2. The above mentioned NCMC's shall not be considered for vertical progression as well as for the calculation of SGPA/CGPA but completion of the courses shall be mandatory for the award of degree.
3. CIE marks will be allotted by the concerned course teacher based on attendance and performance in the practice sessions/field every semester (III to VI) for 100 marks and marks scored by the student are scaled down to 100 in the VI semester.
4. In case, any student fails to secure the minimum 40% of the prescribed marks, he/she shall be deemed to have secured 'F' grade.

* Lateral Entry Students:

The Non-Credit Mandatory Course, Additional Mathematics I: is prescribed for III Semester Lateral Entry Diploma students admitted to III Semester of BE Program. The student shall register for this course along with other III semester courses. The students shall attend classes for the course during the semester and complete all formalities of attendance and CIE. In case, any student fails to secure the minimum 40% of the prescribed CIE marks, he/she shall be deemed to have secured 'F' grade. In such a case, the student has to fulfill the requirements during subsequent semester/s to appear for CIE. In case student fails to register for the said course/ falls short of attendance, he/she will repeat the course whenever it is offered next. Additional Mathematics I shall

have CIE component only and no SEE component. This Course shall not be considered for vertical progression, but completion of the course shall be mandatory for the award of the degree.

Lateral Entry Students:

The Non-Credit Mandatory Course, Inter/Intra Institutional Internship: All the students admitted under lateral entry category shall have to undergo a mandatory Internship of 02 weeks during the intervening vacation of III and IV semesters. Summer Internship shall include Inter / Intra Institutional activities. A Viva-voce examination shall be conducted during IV semester based on which a student will be declared pass or fail. The internship shall be considered as a head of passing for the award of degree. Those, who do not take up / complete the internship shall be declared fail and shall have to complete internship requirements during subsequent semesters.

Lateral Entry Students:

AICTE Activity Points Requirement for BE Programs: As per AICTE guidelines (refer Chapter 6 – *AICTE Activity Point Program, Model Internship Guidelines*), in addition to academic requirements, students must earn a specified number of **Activity Points** to be eligible for the award of the degree. The points to be earned is:

1. **Regular students** admitted to a 4-year degree program must earn **100 Activity Points**.
2. **Lateral entry students** (joining from the second year) must earn **75 Activity Points**.
3. **Students transferred** from other universities directly into the V semester must earn **50 Activity Points** from the date of entry into VTU.

These Activity Points are non-credit and will not be considered for the SGPA/CGPA or be used for vertical progression. However, earning Activity Points is mandatory for the award of the degree, and the points earned will be reflected on the eighth semester Grade Card. If a student completes all the semesters (eight or six) at the end of the program but fails to earn the required Activity Points, the VIII semester Grade Card will be withheld until the requirement is fulfilled. Also, the degree will be awarded only after the Grade Card has been released. The hours spent earning the activity points will not be counted for regular attendance requirements. Students can accumulate these points at any time during their program period, including weekends, holidays, and vacations, starting from the year of admission, provided they meet the minimum hours of engagement prescribed for each activity by AICTE.

B.E. in Information Science and Engineering
SCHEME OF TEACHING III SEMESTER Batch:2024

IV SEMESTER

Sl. No.	Course Code	Course Name	Teaching Department	Category	Credits				Total contact hours /week			
					L	T	P	Total	L	T	P	S*
1.	IS41	Statistics, Probability and Linear Programming	Maths	ASC	2	1	0	3	2	2	0	2
2.	IS42	Microcontrollers	ISE	PCC(IC)	3	0	1	4	3	0	2	3
3.	IS43	Design and Analysis of Algorithms	ISE	PCC	3	0	0	3	3	0	0	3
4.	IS44	Database Management Systems	ISE	PCC	3	0	0	3	3	0	0	3
5.	IS45	Advanced Java	ISE	PCC	3	0	0	3	3	0	0	3
6.	ISL46	Design and Analysis of Algorithms Lab	ISE	PCC	0	0	1	1	0	0	2	0
7.	ISL47	Database Management Systems Lab	ISE	PCC	0	0	1	1	0	0	2	0
8.	ISL48	Advanced Java Lab	ISE	PCC	0	0	1	1	0	0	2	0
9.	ISAEC49X	Ability Enhancement Course - IV	ISE	AEC	1	0	0	1	1	0	0	1
				Total	15	1	4	20	15	2	6	15
10.	INT410	Inter/ Intra Institutional Internship	ISE	NCMC	0	0	0	0				
11.	AM41	Additional Mathematics II *	Maths	NCMC	0	0	0	0	3			

Ability Enhancement Course- IV

S.No.	Course Code	Course Name
1.	ISAEC491	UI/UX
2.	ISAEC492	Data Analytics with R Programming

Nomenclature: **ASC:** Applied Science Course, **IC:** Integrated Course (Practical Integrated with Theory Course), **PCC:** Professional Core Course, **AEC:** Ability Enhancement Course, **NCMC:** Non-credit Mandatory Course

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The Program Core Course(IC) is an Integrated course, combining theory with practical components.

- Theory sessions shall be conducted for 3 hours per week, while the practical sessions shall be conducted for 2 hours per week.
- Theory components shall be evaluated through both Continuous Internal Evaluation (CIE) and Semester End Examination (SEE).
- The practical component shall be assessed only through CIE (No SEE).
- However, questions from the practical component of the Integrated course (IC) can be included in the SEE question paper.

*** Lateral Entry Students:**

The Non-Credit Mandatory Course, Additional Mathematics II: is prescribed for IV Semester Lateral Entry Diploma students admitted to III Semester of BE Program. The student shall register for this course along with other IV semester courses. The students shall attend classes for the course during the semester and complete all formalities of attendance and CIE. In case, any student fails to secure the minimum 40% of the prescribed CIE marks, he/she shall be deemed to have secured 'F' grade. In such a case, the student has to fulfill the requirements during subsequent semester/s to appear for CIE. Incase student fails to register for the said course/ falls short of attendance, he/she will repeat the course whenever it is offered next. Additional Mathematics II shall have CIE component only and no SEE component. This Course shall not be considered for vertical progression, but completion of the course shall be mandatory for the award of the degree.

Lateral Entry Students:

The Non-Credit Mandatory Course, Inter/Intra Institutional Internship: All the students admitted under lateral entry category shall have to undergo a mandatory Internship of 02 weeks during the intervening vacation of III and IV semesters. Summer Internship shall include Inter / Intra Institutional activities. A Viva-voce examination shall be conducted during IV semester based on which a student will be declared pass or fail. The internship shall be considered as a head of passing for the award of degree. Those, who do not take up / complete the internship shall be declared fail and shall have to complete internship requirements during subsequent semesters.

Lateral Entry Students:

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2. **Lateral entry students** (joining from the second year) must earn **75 Activity Point**.3. **Students transferred** from other universities directly into the V semester must earn **50 Activity Points** from the date of entry into VTU.

These Activity Points are non-credit and will not be considered for the SGPA/CGPA or be used for vertical progression. However, earning Activity Points is mandatory for the award of the degree, and the points earned will be reflected on the eighth semester Grade Card. If a student completes all the semesters (eight or six) at the end of the program but fails to earn the required Activity Points, the VIII semester Grade Card will be withheld until the requirement is fulfilled. Also, the degree will be awarded only after the Grade Card has been released. The hours spent earning the activity points will not be counted for regular attendance requirements. Students can accumulate these points at any time during their program period, including weekends, holidays, and vacations, starting from the year of admission, provided they meet the minimum hours of engagement prescribed for each activity by AICTE.

III SEMESTER

Laplace Transforms & Vector Space	
Course Code: IS31	Credits: 2:1:0
Pre-requisites: Calculus and Basics of Linear Algebra	Contact Hours: 30L+30T+30S
Course Coordinator: Dr. Govindaraju M V	

Course Content

Unit I

Laplace Transforms: Definition, transforms of elementary functions, properties of Laplace transforms, existence conditions, transform of derivatives, integrals, multiplication by t^n , division by t , evaluation of integrals by Laplace transforms, transform of periodic function.

- Pedagogy/Course delivery tools: ➤ Chalk and talk, PowerPoint Presentation, Videos
- Links: ➤ <https://nptel.ac.in/courses/111/105/111105134/>
➤ <https://nptel.ac.in/courses/111/105/111105035/>
- Impartus recording: ➤ <https://a.impartus.com/ilc/#/course/119640/593>
➤ <https://a.impartus.com/ilc/#/course/59742/295>

Unit II

Application of Laplace Transforms: Unit-step function, unit-impulse function. inverse transforms, convolution theorem, solution of linear differential equations and simultaneous linear differential equations using Laplace transforms, engineering applications.

- Pedagogy/Course delivery tools: ➤ Chalk and talk, Power Point Presentation, Videos
- Links: ➤ <https://nptel.ac.in/courses/111/105/111105134/>
➤ <https://nptel.ac.in/courses/111/105/111105035/>
- Impartus recording: ➤ <https://a.impartus.com/ilc/#/course/119640/593>
➤ <https://a.impartus.com/ilc/#/course/59742/295>

Unit III

Vector space and Linear Transformation: Vector space, linear combination and span, linearly independent and dependent vectors, basis and dimension, linear transformations, matrix of transformations, rotation about the origin, dilation, contraction and reflection, composition of matrix transformations, kernel and range, change of basis.

- Pedagogy/Course delivery tools: ➤ Chalk and talk, Power Point Presentation, Videos
- Links: ➤ <https://nptel.ac.in/courses/111/105/111105035/>
➤ <https://nptel.ac.in/courses/111/102/111102152/>

- Impartus recording: ➤ <https://a.impartus.com/ilc/#/course/621524/1030>
- <https://a.impartus.com/ilc/#/course/619570/1030>

Unit IV

Orthogonal Projections: The null space of A, solving $Ax = 0$ and $Rx = 0$, the complete solution to $Ax = b$, dimensions of the four subspaces, orthogonality of the four subspaces, projections. orthonormal bases and Gram-Schmidt method, QR-factorization, least-squares approximations.

- Pedagogy/Course delivery tools: ➤ Chalk and talk, Power Point Presentation, Videos
- Links: ➤ <https://nptel.ac.in/courses/111/105/111105035/>
- <https://nptel.ac.in/courses/111/102/111102152/>
- Impartus recording: ➤ <https://a.impartus.com/ilc/#/course/621524/1030>
- <https://a.impartus.com/ilc/#/course/619570/1030>

Unit V

Applications of Eigenvalue Decomposition: Introduction to eigenvalues and eigenvectors, similarity and diagonalization. symmetric matrices, complex matrices, Hermitian and unitary matrices, positive definite matrices, the singular value decomposition (SVD), principal component analysis (PCA), applications to linear recurrence relations, quadratic forms and conic sections.

- Pedagogy/Course delivery tools: ➤ Chalk and talk, PowerPoint Presentation, Videos
- Links: ➤ <https://nptel.ac.in/courses/111/105/111105035/>
- <https://nptel.ac.in/courses/111/102/111102152/>
- Impartus recording: ➤ <https://a.impartus.com/ilc/#/course/621524/1030>
- <https://a.impartus.com/ilc/#/course/619570/1030>

Text Books:

1. B.S.Grewal - Higher Engineering Mathematics - Khanna Publishers – 44th edition-2017.
2. David C. Lay, Steven R. Lay and Judi J. Mc. Donald – Linear Algebra and its Applications – Pearson – 5th edition – 2015.
3. Gilbert Strang, Linear Algebra and its Applications, 5th Edition (2016).

Reference Books:

1. Peter V. O'Neil – Advanced Engineering Mathematics – Cengage learning – 7th edition – 2011.
2. Gareth Williams – Linear Algebra with Applications, Jones and Bartlett Press – 9th edition – 2017.
3. Erwin Kreyszig-Advanced Engineering Mathematics-Wiley-India publishers- 10th edition-2015.

Self-learning:

Problem solving and Quiz.

Course Outcomes(COs):

At the end of the course, students will be able to:

1. Evaluate Laplace Transforms of given function and understand their properties (PO-1, 2 & PSO-1, 3)
2. Obtain inverse Laplace transforms and use it to solve system of ODE's (PO-1, 2 & PSO-1, 3)
3. Obtain matrix of linear transformation. (PO-1, 2 & PSO-1, 3)
4. Solve the system of equations by Least-Squares method. (PO-1, 2 & PSO-1, 3)
5. Obtain eigenvalue decomposition of a matrix and use it to study the concepts of SVD and PCA. (PO-1, 2 & PSO-1, 3)

Course Assessment and Evaluation:

Continuous Internal Evaluation: 50 Marks		
Assessment Tool	Marks	Course outcomes addressed
Internal test-I	30	CO1, CO2, CO3
Internal test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other components	Marks	Course outcomes addressed
Quiz	10	CO1, CO2, CO3
Assignment	10	CO3, CO4, CO5
Semester End Examination:	100	CO1, CO2, CO3, CO4, CO5

Digital Design and Computer Organization	
Course Code: IS32	Credits: 3:0:1
Pre – requisite: Basic Electronics	Contact Hours: 45L+30P+45S
Course Coordinator: Dr. Anitha K	

Course Content

Unit I

Introduction to Digital Design: Introduction to binary logic, basic theorems, Boolean functions and simplification. The Map Method for simplifying Boolean expressions, Two, Three and Four-Variable Map, Don't-Care Conditions, NAND and NOR Implementation, Exclusive-OR Function, Hardware Description Language – Verilog code for simple circuits.

- Pedagogy/Course Delivery tools: Chalk and talk, Power Point Presentation
- Link: https://onlinecourses.nptel.ac.in/noc24_cs61
- Lab Session: Verilog codes for combinational logic circuits

Unit II

Combinational Logic: Introduction, Design Procedure, Binary Adders-subtractor, Decoders, Encoders, Multiplexers and Demultiplexers. HDL for Combinational Circuits.

Sequential Logic: Introduction, Sequential Circuits, Storage Elements: SR, JK, D and T Flip-Flops (only block diagram approach), Function tables and Characteristics equation of all 4 FFs. Mod-8 Asynchronous counter using T FF and Mod-5 Synchronous Counter with design. HDL for Sequential logic Circuits.

- Pedagogy/Course Delivery tools: Chalk and talk, Power Point Presentation
- Link: <https://edurev.in/p/155137/Combinational-Circuits>
- Link: <https://dec-iitkgp.vlabs.ac.in/exp/sequential-circuits/theory.html>
- Lab Session: Verilog program for Adder and Subtractor.
Verilog program for Decoders and Encoders.

Unit III

Basic Structure of Computers: Functional Units, Basic Operational Concepts, Bus structure, Performance – Processor Clock, Basic Performance Equation, Clock Rate, Performance Measurement. Machine Instructions and Programs: Memory Location and Addresses, Memory Operations, Instructions and Instruction sequencing, Addressing Modes.

- Pedagogy/Course Delivery tools: Chalk and talk, Power Point Presentation
- Link: https://onlinecourses.nptel.ac.in/noc20_cs64
- Lab Session: Instruction Execution using Marie Simulator.
ALU design using Logisim simulator.

Unit IV

Input/output Organization: Accessing I/O Devices, Interrupts – Interrupt Hardware, Enabling and Disabling Interrupts, Handling Multiple Devices, Direct Memory Access: Bus Arbitration, The memory System: Basic Concepts, Speed, size and Cost of memory systems. Cache Memories – Mapping Functions.

- Pedagogy/Course Delivery tools: Chalk and talk, Power Point Presentation
- Link: https://onlinecourses.nptel.ac.in/noc20_cs64
- Lab Session: Memory operation using Logisim simulator.
-

Unit V

Basic Processing Unit: Some Fundamental Concepts: Register Transfers, Performing ALU operations, fetching a word from Memory, Storing a word in memory. Execution of a Complete Instruction. Pipelining: Basic concepts, Role of Cache memory, Pipeline Performance.

- Pedagogy/Course Delivery tools: Chalk and talk, Power Point Presentation
- Link: https://onlinecourses.nptel.ac.in/noc20_cs64
- Lab Session: Pipelining demonstration using CPU-OS simulator.

List of Simulation Programs:

1. Verilog codes for combinational logic circuits.
2. Verilog program for Adder and Subtractor.
3. Verilog program for Decoders and Encoders.
4. Verilog program for Multiplexer and Demultiplexer.
5. Verilog program for flip flops.
6. Instruction Execution using Marie Simulator.
7. ALU design using Logisim simulator.
8. Memory operation using Logisim simulator.
9. Pipelining demonstration using CPU-OS simulator.

Text Books:

1. M. Morris Mano & Michael D. Ciletti, Digital Design with an Introduction to Verilog Design, 6e, Pearson Education, 2018.
2. Carl Hamacher, Zvonko Vranesic, Safwat Zaky, Computer Organization, 5th Edition, Tata McGraw Hill, 2002

Reference Books:

1. Donald P Leach, Albert Paul Malvino and Goutam Saha, Digital Principles and Applications, 8th Edition, 2024, Tata McGraw Hill.
2. Stephen Brown, Zvonko Vranesic, Digital Logic Design with VHDL, Third Edition, 2008, Tata McGraw Hill.

Course outcomes (COs):

At the end of the course, the student will be able to:

1. Apply various simplification techniques to realize digital circuits using basic and universal gates. (PO1,2,3,5,9,10 & PSO2)
2. Design different types of combinational and sequential circuits to develop digital system. (PO1,2,3,4,5,9,10,PSO2)
3. Describe the fundamentals of machine instructions, addressing modes to evaluate Processor performance. (PO1 & PSO2)
4. Understand the various methods used for communication between processor and I/O devices for data and control flow. (PO1 & PSO2)
5. Analyse internal Organization of Main Memory/Cache to evaluate Processor Performance. (PO1,2 & PSO2)

Web links and Video Lectures (e-Resources):

- <https://nptel.ac.in/courses/106105185>

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

- Coding Demo
- S* Self-learning-S

Unit No.	Assignment Problems
I	For the following scenarios build the logic circuits (Separately) 1. An AI chatbot adjusts its response tone based on user mood and time of day. 2. An AI sensor system monitors four safe conditions 3. Build an alert system using NOR gates 4. An AI compares two 4 bit patterns for image recognition
II	For the following scenarios build the logic circuits (Separately) 1. An AI selects one of the AI decision model based on weather conditions 2. An AI robot receives 3 bit code input to select any one of its 6 actions 3. An AI system generates compact binary code for multiple signal inputs 4. Build a digital system that samples a data every 8 seconds 5. AI scheduler rotates between its 5 tasks in cycle
III	Calculate performance 1. An edge device runs a ML model for real-time object detection 2. Two processors run the same ML model but have different clock rates
IV	Design a block diagram 1. An embedded AI device receives inputs from camera and microphone at the same time 2. An LLM deployed on a multi-level memory accesses matrix data and user input data frequently from the memory (Hint use Cache memory)
V	Write the register transfer sequence when: A CPU executes a matrix multiplication for a single layer of a neural network.

Course Assessment and Evaluation:

Continuous Internal Evaluation: 50 Marks		
Assessment Tool	Marks	Course Outcomes addressed
Internal test-I	30	CO1, CO2, CO3
Internal test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other components	Marks	Course Outcomes addressed
Lab programs, Weekly Evaluation and Record	10	CO1, CO2, CO3, CO4, CO5
Lab test	10	CO1, CO2, CO3, CO4, CO5
Semester End Examination	100	CO1, CO2, CO3, CO4, CO5

Data Structures	
Course Code: IS33	Credits: 3:0:0
Pre – requisites: Fundamentals of Computing	Contact Hours: 45L+45S
Course Coordinator: Dr. Kusuma S	

Course Content

Unit I

Introduction to Data Structures: Definition, Classification, Pointers and Dynamic memory allocation, Arrays, Dynamic Allocated Arrays, Structures, Polynomial.

The Stack: Definition, Representation, ADT, Basic operations of Stack and its implementation, Stacks using Dynamic Arrays. Applications of stack: Conversion from Infix to Postfix, Evaluation of Postfix expression.

- **Pedagogy/Course Delivery tools:** Chalk-and-talk and Multimedia presentations, and Code walk-through sessions.
- **Links:** <https://nptel.ac.in/courses/106106133>,
<https://nptel.ac.in/courses/106102064>
<https://nptel.ac.in/courses/106106127>
- **Impartus recording:** <https://a.impartus.com/ilc/#/course/82972/295>

Unit II

Recursion: Definition, Process and programming examples (Tower of Hanoi, Factorial, binary search), Role of Stack in recursion(factorial and Fibonacci).

Queues: Definition, Representation, ADT, Primitive operations of queue and its implementation; Circular queues using Dynamic Arrays, Single and Double-Ended Priority queues, Multiple Stacks and Queues.

- **Pedagogy/Course Delivery tools:** Chalk-and-talk and Multimedia presentations, and Code walk-through sessions.
- **Links:** <https://nptel.ac.in/courses/106106133>,
<https://nptel.ac.in/courses/106102064>
<https://nptel.ac.in/courses/106106127>
- **Impartus recording:** <https://a.impartus.com/ilc/#/course/82972/295>

Unit III

Linked List: Singly Linked Lists and chains, Representing Chains in C, Linked List Stacks and Queues, Polynomial representation and adding polynomials, operations for chain, Circular Linked List , operations for circular linked list, Doubly Linked Lists.

- **Pedagogy/Course Delivery tools:** Chalk-and-talk and Multimedia presentations, and Code walk-through sessions.
- **Links:** <https://nptel.ac.in/courses/106106133>,
<https://nptel.ac.in/courses/106102064>
<https://nptel.ac.in/courses/106106127>
- **Impartus recording:** <https://a.impartus.com/ilc/#/course/82972/295>

Unit IV

Trees: Introduction, Binary Trees, Binary Tree Traversals, Binary Search Tree- Definition, searching, insertion, deletion, joining and splitting, height of a binary search tree

Graphs: Introduction, Definition and Graph Representations.

- **Pedagogy/Course Delivery tools:** Chalk-and-talk and Multimedia presentations, and Code walk-through sessions.
- **Links:** <https://nptel.ac.in/courses/106106133>,
<https://nptel.ac.in/courses/106102064>
<https://nptel.ac.in/courses/106106127>
- **Impartus recording:** <https://a.impartus.com/ilc/#/course/82972/295>

Unit V

Hashing: Introduction, Static Hashing, Dynamic Hashing

Multiway Search Trees: B-Trees :Definition, properties, number of elements in B-Tree, insertion and deletion. **B+ Trees:** Definition, searching, insertion and deletion

- **Pedagogy/Course Delivery tools:** Chalk-and-talk and Multimedia presentations, and Code walk-through sessions.
- **Links:** <https://nptel.ac.in/courses/106106133>,
<https://nptel.ac.in/courses/106102064>
<https://nptel.ac.in/courses/106106127>
- **Impartus recording:** <https://a.impartus.com/ilc/#/course/82972/295>

Text Book:

1. Fundamentals of Data Structures in C by Ellis Horowitz, Sartaj Sahni, and Susan Anderson-Freed, 2nd Edition, Universities Press, 2023, ISBN: 9788173716058

Reference Books:

1. Aaron M. Tanenbaum, Yedidyah Langsam and Moshe J. Augenstein, “Data Structures Using C”, 2nd Edition, PHI, 2009.
2. Michael J. Folk, Bill Zoellick and Greg Riccardi, “File Structures-An Object Oriented Approach with C++”, Pearson Education, 2004
3. Seymour Lipschutz, Data Structures Schaum's Outlines, Revised 1st Ed, McGraw Hill, 2014.
4. Gilberg & Forouzan, Data Structures: A Pseudo-code approach with C, 2nd Ed, Cengage Learning, 2014.
5. Reema Thareja, “Data Structures using C”, 3rd Ed, Oxford press, 2012.
6. Jean-Paul Tremblay & Paul G. Sorenson, An Introduction to Data Structures with Applications, 2nd Ed, McGraw Hill, 2013

Self-Learning - GATE oriented aptitude test, Programming Assignment

Course Outcomes(COs):

At the end of the course, the student will be able to:

1. Apply LIFO principle for stack related problems (PO1, PO2, PO3, PO9, PO10 & PSO1)
2. Analyse the importance of queue data structure and its types for solving real-world problems. (PO1, PO2, PO3, PO9, PO10 & PSO1)
3. Design an effective solution for a given problem using linked list. (PO1, PO2, PO3, PO8, PO9, PO10, PO11 & PSO1)
4. Apply nonlinear data structures to perform various operations on data (PO1, PO2, PO3, PO8, PO9, PO10, PO11 & PSO1)
5. Apply hashing techniques for efficient storage and retrieval of data. (PO1, PO2, PO3, PO8, PO9, PO10, PO11 & PSO1)

Course Assessment and Evaluation :

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test	20	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Gate Based Aptitude Test	10	CO1, CO2, CO3, CO4, CO5
Programming Assignment	10	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

Operating System	
Course Code: IS34	Credits: 3:0:0
Pre – requisites: NIL	Contact Hours: 45L+45S
Course Coordinator: Dr.Jagadeesh Sai D	

Course Contents

Unit 1

Operating Systems Overview: Operating System Structure, Operating System Operations, Process Management, Memory Management, Storage Management, Protection and Security. Process Concept, Process Scheduling, Operations on Processes, Inter-process Communication, Process Scheduling: Basic Concepts, Scheduling Criteria -Scheduling Algorithms, Threads.

Pedagogy: Conceptual Learning using NPTEL video lectures and problem-solving sessions

Links: <https://archive.nptel.ac.in/courses/106/105/106105214/>.

Unit 2

Synchronization: Background, The Critical-Section Problem and its solutions, Synchronization Scheduling Algorithms, Semaphores Classic Problems of Synchronization, Deadlocks: System Model, Deadlock Characterization, Methods for Handling Deadlocks, Deadlock Prevention, Deadlock Avoidance, Deadlock detection, Recovery from deadlock.

Pedagogy: Analytical Learning through NPTEL videos, illustrative case problems, and algorithm tracing

Links: <https://archive.nptel.ac.in/courses/106/105/106105214/>

Unit 3

Memory-Management Strategies: Background, Swapping, Contiguous Memory Allocation, Segmentation, Paging, Structure of the Page Table. Virtual-Memory Management: Background, Demand Paging, Copy-on-Write, Page Replacement, Allocation of Frames.

Pedagogy: Visualization-based Learning with animated NPTEL videos and hands-on paging examples.

Links: <https://archive.nptel.ac.in/courses/106/105/106105214/>

Unit 4

File System: File Concept, Access Methods, Directory and Disk Structure, File System Mounting, File Sharing, Implementing File Systems: File, System Structure, File-System Implementation, Allocation Methods, Free space Management.

Pedagogy: Visualization-based Learning with animated NPTEL videos and hands-on paging examples

Links: <https://archive.nptel.ac.in/courses/106/105/106105214/>

Unit 5

Mass Storage Structure: Overview, Disk Structure, Disk Attachment, Disk Scheduling. System Protection: Goals of Protection, Principles of Protection, Domain of Protection, Access Matrix, Implementation of Access Matrix, Access Control.

Pedagogy: Real-world Case-based Learning with NPTEL content and practical protection scenario discussions

Links: <https://archive.nptel.ac.in/courses/106/102/106102132/>

Text Books:

1. Operating System Concepts, 10th Edition by Abraham Silberschatz, Greg Gagne, Peter B. Galvin. 2022 Wiley Publications.

Reference Books:

1. Modern Operating Systems, 4th Edition by Andrew S. Tanenbaum, Herbert Bos. 2021, Pearson.

2. Operating Systems: Internals and Design Principles, 9th Edition by William Stallings. 2021, Pearson.

Self-Learning: Case Study on Linux

Linux Architecture – Kernel vs Shell, System Boot Process

Process Management in Linux – ps, top, nice, kill, process creation using fork, exec

Synchronization in Linux – Semaphores, inter-process communication (ipcs, ipcrm), race conditions

Memory Management – Use of free, vmstat, /proc/meminfo, paging, virtual memory

File System Structure – Linux file hierarchy, df, du, permissions (chmod, chown, setfacl)

Storage Management – Disk scheduling via iostat, partitioning with fdisk, filesystem mounting

Security & Protection – User/group management, access control via ACLs (getfacl, setfacl).

Course Outcomes:

CO1: Understand the structure and functionality of the operating system and apply CPU scheduling algorithms for the given problem. (PO1, PO2, PO3, PO9, PO10, PO11, PSO2)

CO2. Analyze the various algorithms for process synchronization and deadlock handling. (PO1, PO2, PSO2)

CO3. Apply the various techniques for memory management. (PO1, PO2, PO3, PO9, PO10, PO11, PSO2)

CO4. Understand file and secondary storage management strategies. (PO1, PO2, PO9, PO10, PO11, PSO2)

CO5. Describe the need for information protection mechanisms and the working of modern operating system (PO1, PSO2)

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Quiz	20	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

Discrete Mathematical Structures	
Course Code: IS35	Credits: 2:1:0
Pre – requisites: Nil	Contact Hours: 30L + 30T+30S
Course Coordinator: Mrs. Prathima M. N.	

Course Content

Unit I

Fundamentals: Sets and subsets, operations on sets, Sequences. Logic: Propositions and Logical Operations, Conditional statements, Methods of proof.

- **Pedagogy/Course Delivery tools:** Chalk and talk, Power Point Presentation
- **Recommended Resources / Links:**
<https://nptel.ac.in/courses/106106094>
<https://a.impartus.com/ilc/#/course/107790/533>

Unit II

Relations and Digraphs: Product sets and partitions, Relations and digraphs, Paths in relations and digraphs, Properties of relations, Equivalence relations, Operations on relations, Transitive closure and Warshall's algorithm.

- **Pedagogy/Course Delivery tools:** Chalk and talk, Power Point Presentation
- **Recommended Resources / Links:**
<https://nptel.ac.in/courses/106106094>
<https://a.impartus.com/ilc/#/course/107790/533>

Unit III

Functions: Types of Functions, Functions for computer science, Permutation functions, POSETS: Order relations and structures: Extremal elements of partially ordered sets, Lattices.

- **Pedagogy/Course Delivery tools:** Chalk and talk, Power Point Presentation
- **Recommended Resources / Links:**
<https://nptel.ac.in/courses/106106094>
<https://a.impartus.com/ilc/#/course/107790/533>

Unit IV

Graphs: Graphs, Graph terminology and special types of graphs, Representing graphs and graph isomorphism, connectivity, Euler and Hamilton paths.

- **Pedagogy/Course Delivery tools:** Chalk and talk, Power Point Presentation
- **Recommended Resources / Links:**
<https://nptel.ac.in/courses/106106094>
<https://a.impartus.com/ilc/#/course/107790/533>

Unit V

Semigroups and Groups: Binary operations revisited: Tables, Semigroups: Sub semigroup, Sub monoid, Isomorphism, Homomorphism, Groups.

- **Pedagogy/Course Delivery tools:** Chalk and talk, Power Point Presentation
- **Recommended Resources / Links:**
<https://nptel.ac.in/courses/106106094>
<https://a.impartus.com/ilc/#/course/107790/533>

Tutorial Sessions: (Problems will be solved on the following topics)

1. Sets and set operations.
2. Sequence formulas and recognition of regular expressions.
3. Logical operator's application and verification through proof techniques.
4. Relations and their digraph representations.
5. Problems involving operations on relations
6. Symmetric closures, transitive closure, and Warshall's algorithm.
7. Functions and exploring their applications in Computer Science.
8. Partially ordered sets, constructing Hasse diagrams, and topological sorting.
9. Lattices through Hasse diagram methods and their types.
10. Graph terminologies and graphical models.
11. Eulerian and Hamiltonian paths/circuits.
12. Verification of group properties.
13. Semigroups, sub-semigroups and sub-monoids.
14. Isomorphisms and homomorphisms

Text Books:

1. Bernard Kolman, Robert C. Busby, Sharon Cutler Ross, Discrete Mathematical Structures, 6th edition, PHI, 2015.
2. Kenneth H Rosen, Discrete Mathematics and its applications, 8th Edition, Tata McGraw-Hill.2021

Reference Books:

1. Ralph P.Grimaldi, B.V Ramana, Discrete and Combinatorial Mathematics, Fifth edition.
2. J.P.Trembly, R. Manohar, Discrete mathematical structures with applications to Computer Science, McGraw Hill

Self-Learning: Problem Solving and Quiz.

Course Outcomes(COs):

At the end of the course, students will be able to:

1. Apply the concepts of set theory and logical reasoning to verify the correctness of mathematical statements (PO-1,2,9,11 & PSO-1)
2. Analyze the properties and operations of relations in solving various problems. (PO-1,2,9,11 & PSO-1)
3. Understanding the concepts of functions and partially ordered sets for solving the given problem. (PO-1,2,9,11 & PSO-1)
4. Apply the graph theory concepts in solving computational problems. (PO-1,2,9,11 & PSO-1)
5. Apply the concepts of groups for binary operations. (PO-1,2,9,11 & PSO-1)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Quiz/Case Study	10	CO1, CO2, CO3, CO4, CO5
Tutorial	10	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

DATA STRUCTURES LAB	
Course Code: ISL36	Credits: 0:0:1
Pre – requisites: Fundamentals of Computing	Contact Hours: 30P
Course Coordinator: Dr. Kusuma S	

Course Contents

Unit 1

List of Experiments:

1. Write a C program to store and manage a list of records (e.g., patients, students, or employees) using structures, pointers, and dynamic memory allocation. Allow the user to add, view, and search records. Use malloc(), realloc(), and free() for memory management..
2. Write a program to illustrate forward and backward surfing in the web browser using stack(Array implementation). Display the appropriate messages in case of exceptions.
3. Write a program to convert and print a given valid parenthesized infix arithmetic expression to postfix expression. The expression consists of single character operands and binary operators + - * /. Apply the concept of stack data structure to solve this problem.
4. Write a program to evaluate a valid postfix expression using stack. Assume that the postfix expression is read as a single line consisting of non-negative single digit operands and binary operators. The operators are + - * and /.
5. A call center phone system has to hold the phone calls from customers and provide service based on the arrival time of the calls. Write a C program to simulate this system using queue data structure. Program should have options to add and remove the phone calls in appropriate order for their service.
6. Write a program to demonstrate round robin processor scheduling using circular queue (array Implementation) with suitable inputs. Program should have options to add, remove and display Elements of the queue.
7. Write a program to illustrate memory allocation to files based on their size using singly linked list. Program must support the following operations on a singly linked list. a. Inserting a node b. Deleting a node c. Display
8. Design, Develop and Implement a Program in C for the following operations on Singly Circular Linked List (SCLL) with header nodes
 - a. Represent and Evaluate a Polynomial $P(x,y,z) = 6x^2y^2z^4yz^5 + 3x^3yz + 2xy^5z - 2xyz^3$
 - b. Find the sum of two polynomials POLY1(x,y,z) and POLY2(x,y,z) and store the result in POLYSUM(x,y,z). Support the program with appropriate functions for each of the above operations
9. Write a program to illustrate memory allocation to files based on their size using doubly linked list. Program must support the following operations on a doubly linked list. a. Inserting a node b. Deleting a node c. Display
10. Write a program to illustrate forward and backward surfing in the web browser using stack (Linked List implementation). Display the appropriate messages in case of exceptions.

11. Write a program to implement Queue Data Structure (Linked List implementation). Display the appropriate messages in case of exceptions.
12. A hospital database consists of CT images of patients. The physician wants to query the image database and retrieve most similar images which **are** referred as numbers. In order to support this facility implement database indexing using binary search tree and preorder traversal method.

Text Books:

1. Reema Thareja, "Data Structures Using C" 2nd Edition, Oxford University Press, 2014, ISBN: 9780198099307.
2. Yashavant Kanetkar, "Data Structures Through C", 4th Edition (Revised & Updated), BPB Publications, 2022, ISBN: 9789355511898.

Reference Books:

1. Aaron M. Tanenbaum, Yedidya Langsam and Moshe J. Augenstein, "Data Structures Using C", 2nd Edition, PHI, 2009.
2. Michael J. Folk, Bill Zoellick and Greg Riccardi, "File Structures-An Object Oriented Approach with C++", Pearson Education, 2004
3. Semour Lipschutz, Data Structures Schaum's Outlines, Revised 1st Ed, McGraw Hill, 2014.
4. Gilberg & Forouzan, Data Structures: A Pseudo-code approach with C, 2nd Ed, Cengage Learning, 2014.
5. Jean-Paul Tremblay & Paul G. Sorenson, An Introduction to Data Structures with Applications, 2nd Ed, McGraw Hill, 2013

Course Outcomes:

At the end of the course, the student will be able to:

At the end of the course the student will be able to:

1. Design the experiment for the given problem using Data Structures concepts. (PO1, PO2, PO3, PO8, PO9, PSO1)
2. Develop the solution for the given real world problem. (PO1, PO2, PO3, PO4, PO8, PO9, PSO1)
3. Analyze the results and produce substantial written documentation (PO1, PO2, PO4, PO8, PO9, PSO1)

Self-Learning

- Programming assignment through LEET code.

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test	20	CO1, CO2, CO3
Other Components		
Conduction, Record and Viva	30	CO1, CO2, CO3
Semester-End Examination (SEE)	50	CO1, CO2, CO3

Unix System Programming Lab	
Course Code: ISL37	Credits: 0:0:1
Pre – requisites: Nil	Contact Hours: 30P
Course Coordinator: Dr. Yogish H K	

Course Content

Part – A

1. Write a shell program that creates a student database file sdb.txt, which consists of sid, sname, ssem, dept. Display only Student Name, Department, and frequency of occurrences of students belonging to respective departments.
2. Write a shell program which takes two file names as command line arguments, if their contents are the same then remove the second file.
3. Write a shell program that takes a filename as a command line argument and reports whether it is a directory, a regular, a character device, a block device, a link or a pipe file.
4. Write a shell program that accepts one or more file names as arguments and converts all of them to uppercase, provided they exist in the current directory.
5. Write a shell script that accepts two file names as command line arguments, checks if the permissions for these files are identical and if the permissions are identical, output common permission and otherwise output each file name followed by its permissions.
6. Write a shell program to do the followings on strings, find length of a string, whether string is NULL, and two strings are equal or not.
7. Write a shell program to print the command line arguments in reverse order.
8. Write a shell program to print the given number in reversed order.
9. Write a shell program to print the first 25 Fibonacci numbers.
10. Write a shell program to print the prime numbers between the specified range.
11. Write a shell program to search the given key element using linear search.
12. Write a shell program to find the largest of three numbers using a user defined function.

Recommended Resources / Links:

1. NPTEL playlist - Linux Programming & Scripting by Anand Iyer(IIT Madras)
https://www.youtube.com/playlist?list=PLbMVogVj5nJRa3VKt_eyZdJ_DitCz1cvQ
2. NPTEL playlist – Basics of Shell Programming IIT Bombay
<https://www.youtube.com/watch?v=h5Labe97yWM>
https://www.youtube.com/watch?v=rRGCGZ6OHw8&list=PLbMVogVj5nJRa3VKt_eyZdJ_DitCz1cvQ&index=2
3. NPTEL playlist – IISc UNIX and Shell Scripting

PART- B

1. Write a C/C++ program to implement UNIX commands ln, mv, rm commands using APIs.
2. Write a program in C/C++ to display the contents of a named file on the standard output device and write to another file.
3. Write a C/C++ program that reads every 100th byte from the named file, where the file name is given as a command-line argument.
4. Write a C/C++ program to display information of a given file which determines type of file and Inode number, where the file name is given as command line argument.
5. Write a C/C++ program to create a process by using fork () and vfork () system call.
6. Write a C/C++ program to demonstrate the process is Zombie and to avoid the Zombie process.
7. Write a C/C++ program to create and demonstrate Orphan Process.
8. Write a C/C++ program to demonstrate a parent process that uses wait () system call to catch the child's exit code.
9. Write a C/C++ Program to demonstrate race condition.
10. Write a C/C++ program to implement UNIX system () Function, using APIs.
11. Write a C/C++ program to catch, ignore and accept the signal - SIGINT.
12. Write a C/C++ program to create a pipe from the parent to child and send data down the pipe.

Recommended Resources / Links:

<https://archive.nptel.ac.in/courses/106/106/106106144/>

Text Books:

1. Sumitabha Das., Unix Concepts and Applications., 4th Edition., Tata McGraw Hill , 2017
2. W. Richard Stevens: Advanced Programming in the UNIX Environment, 2nd Edition, Pearson Education.

Reference Books:

1. Unix System Programming Using C++ - Terrence Chan, PHI,
2. M.G. Venkatesh Murthy: UNIX & Shell Programming, Pearson.

Course Outcomes (COs):

At the end of the course the student will be able to:

1. Design Shell Scripts to solve given real word modules. (PO1, PO2, PO3, PO9, PO10, PSO1)
2. Apply an appropriate APIs to handle various file related operations. (PO1, PO2, PO3, PO4, PO5, PSO1)
3. Apply an appropriate APIs to manage and control the processes. (PO1, PO2, PO3, PO4, PO5, PSO1)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test	20	CO1, CO2, CO3
Other Components		
Conduction, Calculation, Record and Viva	30	CO1, CO2, CO3
Semester-End Examination (SEE)	50	CO1, CO2, CO3

Universal Human Values	
Course Code: UHV38	Credits: 2:0:0
Pre – requisites: Nil	Contact Hours: 30L+30S
Course Coordinator: Mr. Mushtaq Ahmed D M	

Course Content Unit I

Course Introduction - Need, Basic Guidelines, Content and Process for Value Education

1. Understanding the need, basic guidelines, content and process for Value Education
2. Self-Exploration–what is it? - its content and process; ‘Natural Acceptance’ and Experiential Validation- as the mechanism for self-exploration
3. Continuous Happiness and Prosperity- A look at basic Human Aspirations
4. Right understanding, Relationship and Physical Facilities- the basic requirements for fulfillment of aspirations of every human being with their correct priority
5. Understanding Happiness and Prosperity correctly- A critical appraisal of the current scenario
6. Method to fulfill the above human aspirations: understanding and living in harmony at various levels
 - Pedagogy / Course delivery tools: Chalk and talk, Power point presentation, Videos.
 - Lab component / Practical Topics: Survey/polls for self-exploration
 - Links: Holistic Development and Role of Education <https://youtu.be/sGZtTPe-lhQ>

Unit II

Understanding Harmony in the Human Being - Harmony in Myself!

- Understanding human being as a co-existence of the sentient ‘I’ and the material ‘Body’
- Understanding the needs of Self (‘I’) and ‘Body’ - *Sukh* and *Suvidha*
- Understanding the Body as an instrument of ‘I’ (I being the doer, seer and enjoyer)
- Understanding the characteristics and activities of ‘I’ and harmony in ‘I’
- Understanding the harmony of I with the Body: *Sanyam* and *Swasthya*; correct appraisal of Physical needs, meaning of Prosperity in detail
- Programs to ensure *Sanyam* and *Swasthya*

Practice Exercises and Case Studies will be taken up in Practice Sessions.

- Pedagogy / Course delivery tools: Chalk and talk, Power point presentation, Videos.
- Lab component / Practical Topics: Survey/polls for self-exploration
- Links: Harmony in Human Being- Self and Body
<https://youtu.be/0ERSMkRPQBM>
- Links: Harmony in Human Being- Self <https://youtu.be/83oGJ4oDeIg>
- Links: Harmony between Self and Body Prosperity
https://youtu.be/aJ_BU2OgpKs

Unit III

Understanding Harmony in the Family and Society- Harmony in Human- Human Relationship

1. Understanding Harmony in the family – the basic unit of human interaction
2. Understanding values in human-human relationship; meaning of *Nyaya* and program for its fulfillment to ensure *Ubhay-tripti*;
3. Trust (*Vishwas*) and Respect (*Samman*) as the foundational values of relationship
4. Understanding the meaning of *Vishwas*; Difference between intention and competence
5. Understanding the meaning of *Samman*, Difference between respect and differentiation; the other

salient values in relationship

6. Understanding the harmony in the society (society being an extension of family):
7. Samadhan, Samridhi, Abhay, Sah-astitva as comprehensive Human Goals
8. Visualizing a universal harmonious order in society- Undivided Society (Akhand Samaj), Universal Order (Sarvabhaum Vyawastha)- from family to world family!

Practice Exercises and Case Studies will be taken up in Practice Sessions.

- Pedagogy / Course delivery tools: Chalk and talk, Power point presentation, Videos.
- Lab component / Practical Topics: Survey/polls for self-exploration
- Links: Harmony in Family- Trust <https://youtu.be/F2KVVW4WNnS8>
- Links: Harmony in family- Respect https://youtu.be/iLqNRPuv0_8
- Links: Harmony in family- Other Feeling Justice <https://youtu.be/TcYJB7reKnM>
- Links: Harmony in the Society <https://youtu.be/BkWgFinrnPw>

Unit IV

Understanding Harmony in the Nature and Existence - Whole existence as Co- existence

1. Understanding the harmony in the Nature
2. Interconnectedness and mutual fulfillment among the four orders of nature- recyclability and self-regulation in nature
3. Understanding Existence as Co-existence (*Sah-astitva*) of mutually interacting units in all-pervasive space
4. Holistic perception of harmony at all levels of existence

Practice Exercises and Case Studies will be taken up in Practice Sessions.

- Pedagogy / Course delivery tools: Chalk and talk, Power point presentation, Videos.
- Lab component / Practical Topics: Survey/polls for self-exploration
- Links: Harmony in Nature https://youtu.be/K1Jpd_ojydw
- Links: Harmony in Existence https://youtu.be/mormUeZ_RUE

Unit V

Implications of the above Holistic Understanding of Harmony on Professional Ethics

- Natural acceptance of human values
- Definitiveness of Ethical Human Conduct
- Basis for Humanistic Education, Humanistic Constitution and Humanistic Universal Order
- Competence in professional ethics:
 - a. Ability to utilize the professional competence for augmenting universal human order
 - b. Ability to identify the scope and characteristics of people-friendly and eco- friendly production systems,
 - c. Ability to identify and develop appropriate technologies and management patterns for above production systems.
- Case studies of typical holistic technologies, management models and production systems
- Strategy for transition from the present state to Universal Human Order:
 - a. At the level of individual: as socially and ecologically responsible engineers, technologists and managers
 - b. At the level of society: as mutually enriching institutions and organizations

Suggested Learning Resources:

- Pedagogy / Course delivery tools: Chalk and talk, Power point presentation, Videos.
- Lab component / Practical Topics: Survey/polls for self-exploration

Text Books:

1. **R.R Gaur, R Sangal, G P Bagaria**, A foundation course in Human Values and professional Ethics, Excel books, New Delhi, 2010, ISBN 978-8-174-46781-2

Reference Books:

1. **B L Bajpai**, 2004, Indian Ethos and Modern Management, New Royal Book Co., Lucknow. Reprinted 2008.
2. **PL Dhar, RR Gaur**, 1990, Science and Humanism, Commonwealth Publishers.
3. **Sussan George**, 1976, How the Other Half Dies, Penguin Press. Reprinted 1986, 1991
4. **Ivan Illich**, 1974, Energy & Equity, The Trinity Press, Worcester, and HarperCollins, USA
5. **Donella H. Meadows, Dennis L. Meadows, Jorgen Randers**, William W. Behrens III, 1972, limits to Growth, Club of Rome's Report, Universe Books.
6. **Subhas Palekar**, 2000, How to practice Natural Farming, Pracheen(Vaidik) Krishi Tantra Shodh, Amravati.
7. **A Nagraj**, 1998, Jeevan Vidya ek Parichay, Divya Path Sansthan, Amarkantak.
8. **E.F. Schumacher**, 1973, Small is Beautiful: a study of economics as if people mattered, Blond & Briggs, Britain.
9. **A.N. Tripathy**, 2003, Human Values, New Age International Publishers.

Web links and Video Lectures (e-Resources):

1. https://www.youtube.com/channel/UCQxWr5QB_eZUnwxSwxXEkQw
2. <https://www.youtube.com/watch?v=P4vjfE-YnVk&list=PLWDeKF97v9SP7wSlapZcQRrT7OH0ZIGC4>
3. **Course handouts:** https://drive.google.com/drive/folders/1zioX_4L2fCNX4Agw282PN86pcZZT3Osr?usp=sharing
4. **Presentation slides:** https://drive.google.com/drive/folders/1rMUKh1s0HPRBlpp_b1mpS-duNRcwS6YH?usp=sharing

Course Outcomes (COs):

At the end of the course, Students will be able to:

1. Apprehend the need of Value Education over Human aspirations (PO-6)
2. Assimilate Harmony over the physical needs and to overcome the self- needs for a prosperous life. (PO-6)
3. Recognize the need of Harmony in the Family and Society for a better World. (PO-6)
4. Explain the need of mutual understanding for Holistic Harmony in all the Levels of Human Existence. (PO-6)
5. Explain the Holistic understanding of Harmony and Professional Ethics at Individual Level and Society. (PO-6,8)

Course Assessment and Evaluation :

Continuous Internal Evaluation (CIE)		
Assessment Tool	Marks	Course outcomes addressed
Internal test-I	30	CO1, CO2, CO3
Internal test-II	30	CO3, CO4, CO5
Average of the two internal tests will be taken for 30 marks.		
Other components		
• Assignment • Quiz • Presentation • Model / mini project • Any other	20 (10 + 10)	CO1, CO2, CO3, CO4, CO5
Semester End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

Green IT and Sustainability	
Course Code: ISAE391	Credits: 1:0:0
Pre – requisites: Nil	Contact Hours: 15L+15S
Course Coordinator: Dr. Krishna Raj P M	

Course Contents

Unit 1

Introduction: Foundations of sustainable design and HCI, sustainability and usability concerns, scope and objectives of sustainable interaction design.

Pedagogy: Conceptual Learning using NPTEL video lectures and problem-solving sessions

Recommended Resources / Links: <https://nptel.ac.in/courses/106101060>

Unit 2

Usability and Human–Computer Interaction (HCI): Usability principles, evaluation of usability, role of HCI in sustainable design, challenges in achieving usable and sustainable systems.

Pedagogy: Analytical Learning through NPTEL videos, illustrative case problems, and algorithm tracing

Recommended Resources / Links: <https://nptel.ac.in/courses/107103081>

Unit 3

User Participation in the System Development Process: Importance of user involvement, participatory design, sustainability through user-centered design approaches.

Pedagogy: Visualization-based Learning with animated NPTEL videos and examples

Recommended Resources / Links: <https://nptel.ac.in/courses/107103081>

Unit 4

Physical, Cognitive and Affective Engineering; Color, Prototyping and Navigation; Principles and Guidelines, Design Evaluation and Testing; Task Analysis.

Pedagogy: Structure-oriented Learning using NPTEL lectures and implementation exercises

Recommended Resources / Links: <https://nptel.ac.in/courses/107103081>

Unit 5

Models and Methodologies; New Participative Methodology for Sustainable Design (NPMSD); Future ICTs—Present Trends for Future Developments.

Pedagogy: Real-world Case-based Learning with NPTEL content and practical scenario discussions

Recommended Resources / Links: <https://nptel.ac.in/courses/107103081>

Self-Learning:

Case Study: Analyze a company's green IT initiatives, focusing on data center efficiency and e-waste management.

Mini-Project: Propose a "Green Campus" plan by conducting an IT energy audit and suggesting solutions

Text Books:

1. Sustainable Design: HCI, Usability and Environmental Concerns, Tomayess Issa, Pedro Isaias, 2nd Edition, Springer, 2022.

Reference Books:

1. Green Information Technology – A Sustainable Approach, Mohammad Dastbaz, Colin Pattinson, Babak Akhgar, Elsevier, 2015 Inc.

Course Outcomes:

At the end of the course, the student will be able to:

1. Understand the foundational concepts of Green IT and its importance in addressing global environmental challenges.(PO6, PO7, PO8, PSO2)
2. Analyze the environmental impact of IT infrastructure, including data centers, and propose strategies for energy efficiency and resource optimization.. (PO1, PO2, PO4, PSO2)
3. Evaluate sustainable computing practices to minimize the carbon footprint of IT operations. (PO1, PO3, PO5, PSO1)
4. Develop and implement policies for responsible e-waste management and sustainable IT procurement. (PO6, PO8, PO11, PSO3)
5. Apply sustainable software development principles to create energy-efficient and eco-friendly applications.(PO1, PO3, PO5, PSO1)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Case Study Presentation	15	CO1, CO2, CO3, CO4, CO5
MCQ Test	05	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

Capacity Planning for IT	
Course Code: ISAE392	Credits: 1:0:0
Pre – requisites: Nil	Contact Hours: 15L+15S
Course Coordinator: Dr. Krishna Raj P M	

Course Content

Unit I

Goals, Issues, and Processes: capacity planning, Quick and Dirty Math, Predicting When Your Systems Will Fail , Make Your System Stats Tell Stories, Buying Stuff: Procurement Is a Process, Performance and Capacity: Two Different Animals, The Effects of Social Websites and Open APIs

Setting Goals for Capacity: Different Kinds of Requirements and Measurements, Architecture Decisions.

Pedagogy/Course Delivery tools: Chalk and talk, Power Point Presentation

Links: <https://www.youtube.com/watch?v=w0cD26CLBA0>

- <https://www.youtube.com/watch?v=5-hhfBXykec>

Unit II

Measurement: Units of Capacity: Aspects of Capacity Tracking Tools, Applications of Monitoring.

Pedagogy/Course Delivery tools: Chalk and talk, Power Point Presentation

Link: <https://www.youtube.com/watch?v=9e4IohiFmZ8&t=63s>

Unit III

Measurement: API Usage and Its Effect on Capacity, Examples and Reality,

Predicting Trends: Riding Your Waves.

Pedagogy/Course Delivery tools: Chalk and talk, Power Point Presentation

Link: <https://www.youtube.com/watch?v=qj4ziswxupE>

Unit IV

Predicting Trends: Procurement, The Effects of Increasing Capacity , Long-Term Trends, Iteration and Calibration

Deployment: Automated Deployment Philosophies, Automated Installation Tools , Automated Configuration.

Pedagogy/Course Delivery tools: Chalk and talk, Power Point Presentation.

Link: <https://www.youtube.com/watch?v=jTW79ofC6Go>

Unit V

Virtualization and Cloud Computing: Virtualization, Cloud Computing, Computing Resource Evolutions, Mixed Definitions, Cloud Capacity, Use it or lose it (your wallet), Measuring the clouds, Cloud Case Studies, Cloud Use Case: Anonymous Desktop Software Company.

Pedagogy/Course Delivery tools: Chalk and talk, Power Point Presentation

Link: https://www.youtube.com/watch?v=_pPlanX5wQY

Self Learning: Tool demonstration

Text Books:

1. John Allspaw, The Art of Capacity Planning, 2008, O'Reilly

Course Outcomes (COs):

At the end of the course, the student will be able to:

1. Analyze the requirement and measurements for capacity planning by considering the goal, issues, and processes (PO-1,2 ,7,9,10,11,12 & PSO-2)
2. describe capacity measurement and monitoring (PO-1,2,7,9,10,11,12 & PSO-2)
3. Analysis the measurement data for prediction towards overall planning process. (PO-1,2,5,7,9,10,11,12 & PSO-2)
4. understand the concepts related to deployment, installation, configuration, and management. (PO-1,2,5,7,9,10,11,12 & PSO-2)
5. Evaluate how the virtualization and cloud services fit into a capacity plan. (PO- 1,2,7,9,10,11,12 & PSO-2)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Quiz	10	CO1, CO2, CO3, CO4, CO5
Case Study	10	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

Numerical Analysis with Python	
Course Code: ISAE393	Credits: 1:0:0
Pre – requisites: Nil	Contact Hours: 15L+15S
Course Coordinator: Dr. Krishnaraj P M	

Course Content

Unit I

Introduction to Python: Overview of Python, basic syntax, Variables, data types, and operators, Control structures (if statements, loops) and functions.

- **Pedagogy:** Conceptual Learning using NPTEL video lectures and problem-solving sessions
- **Recommended Resources / Links:** <https://nptel.ac.in/courses/106106212>

Unit II

Numpy: Creating Arrays, Indexing and Slicing, Data Type Objects, dtype.

- **Pedagogy:** Analytical Learning through NPTEL videos, illustrative case problems, and algorithm tracing
- **Recommended Resources / Links:** <https://nptel.ac.in/courses/106106212>

Unit III

Numpy Arrays: Numerical Operations on Numpy, Arrays Concatenating, Flattening and Adding Dimensions Boolean Indexing.

- **Pedagogy:** Visualization-based Learning with animated NPTEL videos and hands-on paging examples
- **Recommended Resources / Links:** <https://nptel.ac.in/courses/106106212>

Unit IV

Matplotlib : Line plot, bar plot , histograms, scatter plots, label on axes, multiple plots Pandas: Series Data Structure – Introduction , Indexing, Pandas.Series.Apply

- **Pedagogy:** Structure-oriented Learning using NPTEL lectures and whiteboard implementation exercises
- **Recommended Resources / Links:** <https://nptel.ac.in/courses/106106212>

Unit V

Pandas: Filtering, creating series from dictionaries, Dataframe: need of Dataframes, Dataframes from dictionaries, accessing rows , accessing columns, rearranging, sorting, adding rows and columns.

- **Pedagogy:** Real-world Case-based Learning with NPTEL content and practical scenario discussions
- **Recommended Resources / Links:** <https://nptel.ac.in/courses/106106212>

Self-Learning:

Case Study: Analyze a real-world dataset using Pandas to demonstrate data filtering and analysis techniques.

Mini-Project: Develop a Python script to perform a numerical analysis task, such as plotting a function or solving a system of equations, and visualize the results using Matplotlib.

Text Book:

1. Data Analysis: Numpy, Matplotlib and Pandas, Bernd Klein, 2024.

Reference Book:

1. Automate the boring stuff with python: Practical programming for total beginners, Al Sweigart, 2015.

Course Outcomes (COs):

At the end of the course, the student will be able to:

1. Understand the fundamental concepts of Python programming, including syntax, data types, and control structures, for numerical problem-solving. (PO1, PO2, PO4, PSO1)
2. Apply the NumPy library to perform efficient numerical operations and manipulations on multi-dimensional arrays. (PO1, PO2, PO5, PSO1)
3. Utilize the Matplotlib library to visualize data and the results of numerical analyses through various plot types. (PO1, PO3, PO5, PSO3)
4. Master the Pandas library for effective data handling, filtering, and analysis of structured data. (PO1, PO3, PO5, PSO3)
5. Integrate Python's libraries to solve complex numerical problems and present the findings through clear data visualizations. (PO1, PO3, PO5, PSO1)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Course Project	10	CO1, CO2, CO3, CO4, CO5
Report	10	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

PHYSICAL EDUCATION	
Course Code: PE69	Credits: NCMC
Pre – requisites: Nil	
Course Coordinator: Dr. Kiran Kumar H K	

Course Content

Course Learning Objectives:

1. To introduce students to the importance of physical fitness for success in any career.
2. To instill in them concepts of team spirit and team building
3. To develop positive thinking, goal setting, and decision-making abilities under duress.
4. To harness values and skills like leadership, communication, and sacrifice.
5. To inculcate in students, the ability to handle success and failures with equanimity.

Selection Process: A Student shall select any one of the following Sports based on his/her interest and the facility available. The details of Sports Facilities available (both indoor and outdoor) at the institute campus are as below:

Outdoor Games		Indoor Games		Athletics	
Games	No. of Students	Games	No. of Students	Events	No. of Students
Volleyball	12 x 4 = 48	Badminton	30	Sprint - 100mt, 200mt, 400mt	60
Basketball	12 x 4 = 48	Table Tennis	30	Middle distance running – 800mt, 1,500mt	
Kabaddi	12 x 4 = 48	Chess	30	Long distance running 5,000mt, 10,000mt	
Kho Kho	12 x 4 = 48	Weight Training [Gym]	35	Jumping Events – Long Jump Triple Jump High Jump	30
Throw ball	12 x 4 = 48			Throwing Events Shot Put Discus Javelin	30
Football	16 x 4 = 64	Note: Students should bring their own sports attires			

Contact Sessions: A student shall abide by the following during the sessions scheduled in the semester.

Session 1

Fundamentals of Physical Education, value addition to personality through fitness education, discipline and team building activities, Orientation towards particular sports and skill training

Session 2

Formation of teams based on student's orientation and preference. Team practice and skill enhancement.

Session 3

Conduction of matches in all sporting events registered by Students. Evaluation of each student shall be based on their performance either in team or individual. The student representing the Institute at University/State/National/International Level will be awarded additional marks during evaluation.

Course Outcomes (COs):

1. Develop interest and skill in playing particular sports.
2. Understand the process of organizing sporting events.
3. Appreciate the role of fitness for a better lifestyle.
4. Derive lessons from sports activities for effective planning and discipline in Life.
5. Analyze situations and optimize end results.

Course Assessment & Evaluation:

1. A committee consisting of Sports Director and Coaches of respective Sports will be formed to observe and evaluate the students for CIE in each semester.
2. Students shall follow the schedules, rules and regulations as prescribed by the Committee.
3. Students shall mandatorily have 85% attendance to be eligible for evaluation.
4. All the Sessions and evaluation process will be common for all semesters of the academic year.
5. The final result will be reflected on the grade card of 8th Semester.
6. The final marks shall be calculated after scaling down CIE to 50 marks & combining with 50 marks for SEE.

YOGA	
Course Code: YO69	Credits: NCMC
Pre – requisites: Nil	
Course Coordinator: Dr. Hari Chandra B P & Dr. Parimala P	

Course Content

Course Learning Objectives:

1. To introduce to the students, the fundamental theoretical aspects of yoga.
2. To inculcate in students a habit of practicing yoga.
3. To be able to demonstrate basic yoga asanas.
4. To be able to practice fundamental breathing practices and mudras.
5. To understand the relevance of yoga and research in modern times.

Introduction: Definition of yoga, benefits, astangas of yoga, Relevance of yoga and yoga-research in modern times.

Asanas: Kriyathmakachalanas, Suryanamaskar, Superbrain yoga, Vrikshasana, Trikonasana, Veerabhadrasana, Paschimotasana, Purvothanasana, Bharadwajasana, Amruthasana, Parivruttha Trikonasana, Parsvakonasana, Ustrasana, Padmasana, Jaaanushirshasana, Navasana, Ardachakrasana, Ardhatichakrasana, Jataraparivarthanasana, Sethubandasana, Sarvangasana, Mathyasana, Dhanurasana, Shirshasana. **Pranayamas:** Anuloma-Viloma, Suryanuloma, Chandranuloma, Brahmari, Suryanbedhana, Chandrabedhana, Sheetal, Seethkari, Sadantha, bastrika.

Mudras: Chinmudra-Jnanamudra, Praana mudra, panchaprana mudras, panchabhoota mudras, Pruthvi mudra, Shoonya mudra, Surya mudra, Jalodharanashaka mudra, Kundalini mudra, shoonyavaayu mudra, shakti mudra, sandhi mudra, vajra mudra and garuda mudra.

Course Outcomes (COs):

At the end of the course, a student will

1. Understand the fundamental and theoretical aspects of yoga.
2. Develop a habit of practicing yoga.
3. Demonstrate basic yoga asanas.
4. Demonstrate fundamental breathing practices.
5. Understand the relevance of yoga and its research in modern times.

Reference Books:

1. Light on yoga, B K S Iyengar, Publisher -Thorsons, UK, 2006
2. Light on pranayama, B K S Iyengar, Publisher - Element; First Edition
3. The Essential Yoga Mudras for Healing, Dr. Aasoori K. Rangaraja Iyengar, Saranga Publishing; First Edition 2021

Pedagogy:

Chalk and talk, demonstration, videos, ppt.

Contact Sessions:

There would be one introduction class, and five contact classes in each semester.

The candidates shall practice yoga on a daily basis, or in the worst case on alternate days at their place of residence and maintain a short diary in the format provided by yoga teacher. The same shall be brought to the classes.

Online Reference Sources:

- Yoga for beginners part 1: <https://www.youtube.com/watch?v=VwPeThpwfWI>
- Yoga for beginners part 2: https://www.youtube.com/watch?v=s_pnJTcOp8A
- Suryanamaskar: <https://www.youtube.com/watch?v=nUdlucNd6go&t=133s>
- Yoga for anxiety and stress: https://www.youtube.com/watch?v=hJbRpHZr_d0
- Common yoga protocol: https://www.youtube.com/watch?v=Av5ib_XRKT4
- Relevance of yoga in modern times:
www.youtube.com/watch?v=HUzBCts7BTo

Course Assessment and Evaluation:

1. A committee consisting of Yoga Instructors will be formed to observe and evaluate the students for CIE in each semester.
2. Students shall follow the schedules, rules, and regulations as prescribed by the Committee.
3. Students shall mandatorily have 85% attendance to be eligible for evaluation.
4. All the Sessions and evaluation processes will be common for all semesters of the academic year.
5. The final result will be reflected on the grade card of 8th Semester.
6. The final marks shall be calculated after scaling down CIE to 50 marks & combining it with 50 marks for SEE.

Scheme of SEE

Practical Demonstration	30 marks
Write-up	10 marks
Viva	10 marks
Total	50 marks

NATIONAL SERVICE SCHEME	
Course Code: NS69	Credits: NCMC
Pre – requisites: Nil	
Course Coordinator: Dr. Puttabore Gowda & Dr. Siddaraju C	

Course Content

Course Learning Objectives:

1. To introduce students to the importance of national service
2. To harness values and skills like leadership, teamwork and sacrifice.
3. To serve society through educational services and health
4. To work towards rural and local development through technological services
5. To inculcate in students, the ability to handle socially relevant projects.

Students shall involve in activities related to national and regional technical and non-technical services, as listed below.

- Serving society by bringing awareness on education and cleanliness.
- Blood donation camps
- Developing technologies for rural masses.
- Conduction and participation in camps for a social cause.
- Educating towards health and well-being of individuals/society.
- Cultural and educational programs for society.
- Contributing towards the improvement of civil services and bringing certain shortcomings to the notice of higher authorities for suitable remedial actions.
- Contribution towards traffic management and other public services.
- Clean up and development of water sources around public places.
- Services during a disaster or other needy situations.
- Camps for the rejuvenation of lakes and water bodies.
- Serving nature and agriculture.
- Awareness programs on health and food adulteration.
- Presenting papers/talks in various fora on the above topics.
- Developing technologies for rural masses beyond academic requirements.
- Plantation programs.
- Conducting programs for self-sustainability, and human and national development.
- Contribution towards orphans and challenged individuals through well-recognized organizations.
- Carrying out designated activities in villages.
- Development and implementation of strategies for solid waste, E-waste etc.
- Education towards pollution control and traffic management.
- Production of documentaries and short films/videos for motivating people on any of the above causes.

Course Outcomes (COs):

At the end of the course, a student will be able to

1. Understand the importance of national service.
2. Gain skills like leadership and teamwork.
3. Volunteer towards educational services and health.
4. Contribute to rural and local development through technical services.
5. Comprehend socially relevant projects

Contact Sessions:

The students shall attend the review and contact sessions as scheduled by the course coordinator.

Course Assessment and Evaluation:

1. The candidates shall maintain a record of activities in a Diary, and get them endorsed during the contact sessions at least 3 times in a semester.
2. A detailed project report should be submitted during the last fortnight of the semester.
3. Evaluation will be done during each semester based on the nature of the contribution.
4. The final marks shall be calculated after scaling down CIE to 50 marks & combining with 50 marks for SEE

III Semester
(For Lateral Entry Diploma Students)

ADDITIONAL MATHEMATICS – I

Course Code: AM31

Credits: 0:0:0

Pre-requisites: Nil

Contact Hours: 42

Course coordinators: Dr. Shashi Prabha Gogate S

Unit I

Differential Calculus: Successive differentiation, nth derivatives of some standard functions, Leibnitz theorem, Polar curves. Angle between the radius vector and the tangent, angle between curves, length of the perpendicular from pole to the tangent, pedal equations. Taylor's and Maclaurin's expansions.

- Pedagogy / Course delivery tools: ➤ Chalk and talk
- Links: ➤ <https://nptel.ac.in/courses/111/105/111105121/>
- Impartus recording: ➤ <https://a.impartus.com/ilc/#/course/107625/1030>

Unit II

Integral Calculus: Introduction, Reduction formula, Reduction formula for $\int \sin^n x \, dx$, $\int \cos^n x \, dx$ and $\int \sin^n x \cos^m x \, dx$. Evaluation of double and triple integrals.

- Pedagogy / Course delivery tools: ➤ Chalk and talk
- Links: ➤ <https://nptel.ac.in/courses/111/105/111105121/>
- Impartus recording: ➤ <https://a.impartus.com/ilc/#/course/107625/1030>

Unit III

Vector Algebra: Scalar and vectors. Vector addition and subtraction. Multiplication of vectors (Dot and Cross products). Scalar and vector triple product-simple problems. Vector functions of a single variable. Derivative of a vector function, geometrical interpretation. Velocity and acceleration.

- Pedagogy / Course delivery tools: ➤ Chalk and talk
- Links: ➤ <https://nptel.ac.in/courses/111/105/111105134>
- Impartus recording: ➤ <https://a.impartus.com/ilc/#/course/107625/1030>

Unit IV

Vector Differentiation: Scalar and vector fields, gradient of a scalar field, directional derivative, divergence of a vector field, solenoidal vector, curl of a vector field, irrotational vector. Laplace's operator. Vector identities connected with gradient, divergence and curl.

- Pedagogy / Course delivery tools: ➤ Chalk and talk
- Links: ➤ <https://nptel.ac.in/courses/111/105/111105134>
- Impartus recording: ➤ <https://a.impartus.com/ilc/#/course/107625/1030>

Unit V

First Order Differential Equations: Solution of first order and first degree differential equations, variable separable methods, homogeneous equations, linear and Bernoulli's equations, exact differential equations.

- Pedagogy / Course delivery tools: ➤ Chalk and talk, Power Point Presentation
- Links: ➤ <https://nptel.ac.in/courses/111/105/111105121/>
- Impartus recording: ➤ <https://a.impartus.com/ilc/#/course/59742/295>

Self-learning:

Problem solving and Quiz.

Text Books:

1. **B.S. Grewal** – Higher Engineering Mathematics, Khanna Publishers, 44th edition, 2017.
2. **Erwin Kreyszig** – Advanced Engineering Mathematics – Wiley Publication, 10th Edition, 2015.

Reference Books:

1. **H. K. Dass** – Higher Engineering Mathematics – S Chand Publications, 1998.
2. **B. V. Ramana** – Engineering Mathematics – Tata McGraw-Hill Publishing Co. Ltd., New Delhi, 2008.

Course Outcomes

At the end of the course the student will be able to

1. Solve problems related to nth derivative to some standard functions, polar curves and power series expansions. (PO-1,2)
2. Apply the concept of reduction formula to determine the length, area, volume of revolution of an arc of the curve. (PO-1,2)
3. Solve the problems related to velocity and acceleration. (PO-1,2)
4. Apply vector differentiation to identify solenoidal and irrotational vectors. (PO-1,2)
5. Apply the concept of various methods to solve first order first degree differential equations. (PO-1,2)

Course Assessment and Evaluation:

Continuous Internal Evaluation: 50 Marks		
Assessment Tool	Marks	Course outcomes addressed
Internal test-I	30	CO1, CO2, CO3
Internal test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other components	Marks	Course outcomes addressed
Quiz	10	CO1, CO2, CO3
Assignment	10	CO3, CO4, CO5

IV Semester

Statistics, Probability and Linear Programming	
Course Code: IS41	Credits: 2:1:0
Pre-requisites: Basic Probability	Contact Hours: 30L+30T+30S
Course Coordinators: Dr. Govindaraju M V	

Unit I

Statistics: Curve fitting by the method of least squares, fitting linear, quadratic and geometric curves, correlation, regression and multiple regression.

Probability Distributions: Random variables, Binomial distribution, Poisson distribution.

- Pedagogy/Course delivery tools: ➤ Chalk and talk, Power Point Presentation, Videos
- Links: ➤ <https://nptel.ac.in/courses/111/105/111105035/>
- Impartus recording: ➤ <https://a.impartus.com/ilc/#/course/96127/452>
➤ <https://a.impartus.com/ilc/#/course/132243/636>
➤ <https://a.impartus.com/ilc/#/course/119635/593>

Unit II

Probability Distributions: Uniform distribution, Exponential distribution, Gamma distribution and Normal distribution.

Joint probability distribution: Joint probability distribution (both discrete and continuous), conditional probability, conditional expectation.

- Pedagogy/Course delivery tools: ➤ Chalk and talk, PowerPoint Presentation, Videos
- Links: ➤ <https://nptel.ac.in/courses/111/105/111105035/>
➤ <https://nptel.ac.in/courses/111/107/111107119/>
- Impartus recording: ➤ <https://a.impartus.com/ilc/#/course/96127/452>
➤ <https://a.impartus.com/ilc/#/course/119635/593>

Unit III

Markov Chain: Introduction to stochastic process, probability vectors, stochastic matrices, regular stochastic matrices, Markov chains, higher transition probabilities, stationary distribution of regular Markov chains and absorbing states, Markov and Poisson processes.

Queuing theory: Introduction, symbolic representation of a queuing model, single server Poisson queuing model with infinite capacity ($M/M/1 : \infty /FIFO$), single server Poisson queuing model with finite capacity ($M/M/S : N/FIFO$), multiple server Poisson queuing model with infinite capacity ($M/M/S : \infty /FIFO$), Multiple server Poisson queuing model with finite capacity ($M/M/S : N/FIFO$), introduction to $M/G/1$ queuing model.

- Pedagogy/Course delivery tools: ➤ Chalk and talk, Power Point Presentation, Videos
- Links: ➤ <https://nptel.ac.in/courses/111103022>
- Impartus recording: ➤ <https://a.impartus.com/ilc/#/course/96127/452>

- <https://a.impartus.com/ilc/#/course/119635/593>

Unit IV

Sampling and Statistical Inference: Sampling distributions, central limit theorem, concepts of standard error and confidence interval, level of significance, type I and type II errors, one tailed and two tailed tests, Z-test: for single mean, for single proportion and for difference between means, Student's t –test: for single mean and for difference between two means, F – test for equality of two variances, Chi-square test: for goodness of fit and for independence of attributes.

- Pedagogy/Course delivery tools: ➤ Chalk and talk, Power Point Presentation, Videos
- Links: ➤ <https://nptel.ac.in/courses/111/105/111105035/>
- <https://nptel.ac.in/courses/111/107/111107119/>

Unit V

Linear Programming: Introduction to linear programming problem (LPP), formulation of the problem, graphical method, general, canonical and standard forms of LPP, simplex method, big-M method, two-phase simplex method and duality in linear programming.

- Pedagogy/Course delivery tools: ➤ Chalk and talk, Power Point Presentation, Videos
- Links: ➤ <https://nptel.ac.in/courses/111104027>

Self Learning:

Problem solving and Quiz.

Text Books:

1. R.E. Walpole, R. H. Myers, R. S. L. Myers and K. Ye – Probability and Statistics for Engineers and Scientists – Pearson Education – Delhi – 9th edition – 2012.
2. B.S.Grewal - Higher Engineering Mathematics - Khanna Publishers – 44th edition-2017.
3. T. Veerarajan- Probability, Statistics and Random processes – Tata McGraw-Hill Education – 3rd edition -2017.
1. Kanti Swarup, P.K. Gupta and Man Mohan -Operations Research-Sultan Chand & Sons Publishers– 2014.

Reference Books:

1. Erwin Kreyszig - Advanced Engineering Mathematics-Wiley-India publishers- 10th edition-2015.
2. Sheldon M. Ross – Probability models for Computer Science – Academic Press, Elsevier– 2009.
3. Murray R Spiegel, John Schiller & R. Alu Srinivasan – Probability and Statistics – Schaum's outlines -4th edition-2012.
4. Kishore S. Trivedi – Probability & Statistics with Reliability, Queuing and Computer Science Applications – John Wiley & Sons – 2nd edition – 2008.
5. Johnson/Miller: Miller & Freund's Probability and Statistics for Engineers , Eighth Edition, Pearson Education India -2015

Course Outcomes

At the end of the course, students will be able to

1. Fit a least squares curve to the given data and analyze the given random data and its probability distributions. (PO-1, 2 & PSO-1, 3)

2. Find parameters of Continuous Probability distributions and calculate the marginal and conditional distributions of bivariate random variables. (PO-1, 2 & PSO-1, 3)
3. Predict future events using Markov chain and in queuing models. (PO-1, 2 & PSO-1, 3)
4. Choose an appropriate test of significance and make inference about the population from a sample. (PO-1, 2 & PSO-1, 3)
5. Formulate and solve a simple linear programming problem. (PO-1, 2 & PSO-1, 3)

Course Assessment and Evaluation:

Continuous Internal Evaluation: 50 Marks		
Assessment Tool	Marks	Course outcomes addressed
Internal test-I	30	CO1, CO2, CO3
Internal test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other components	Marks	Course outcomes addressed
Quiz	10	CO1, CO2, CO3
Assignment	10	CO3, CO4, CO5
Semester End Examination:	100	CO1, CO2, CO3, CO4, CO5

MICROCONTROLLERS	
Course Code: IS42	Credits: 3:0:1
Pre – requisites: Computer Organization and Architecture	Contact Hours: 45L + 30P+45S
Course Coordinator: Mr. Prashanth Kambli	

Course Contents

Unit 1

ARM Embedded Systems, RISC design philosophy, ARM design philosophy, Introduction to Embedded systems, Design of Embedded Systems, Applications, Embedded processors, Operating System, Connectivity. Internal Components of System-on-chip, General Microprocessor unit, MCU PIN diagram, Timers and Counters, Pulse width modulator.

Pedagogy: Conceptual Learning using NPTEL video lectures and problem-solving sessions

Recommended Resources / Links:

1. Introduction to Embedded System Design - Course
2. <https://a.impartus.com/ilc/#/video/id/3404203>

Unit 2

Serial Communication, Direct Memory Access, Semiconductor Memory, Designing Low power systems, BUS architecture. Embedded Systems – the Software, Endian-ness, Data Alignment and Memory Banks, Peripheral I/O and Memory Mapped I/O, Load Store Architecture, Stack, FLAGS.

Pedagogy: Analytical Learning through NPTEL videos, illustrative case problems

Recommended Resources / Links:

1. DMA Introduction: Master – Slave Communication | Free Video Tutorial | Udemy
2. <https://a.impartus.com/ilc/#/video/id/3510844>

Unit 3

The Architecture of ARM 7, ARM 7 architecture, Interrupts and Exceptions, ARM7 pipeline, Advanced Features. Assembly Programming of ARM7, Embedded program development, ARM7 Instruction set.

Pedagogy: Assembly programming learning skills and algorithm tracing

Recommended Resources / Links:

1. <https://www.udemy.com/course/armv8-exception-level1/>
2. <https://a.impartus.com/ilc/#/video/id/3542117>

Laboratory Session: Part A 1-5 programs.

Unit 4

Assembly Language Programming, Accessing Memory, Programming of ARM7 using C, ARM7 SOC.

Pedagogy: Assembly programming in high level language learning skills for embedded systems

Recommended Resources / Links:

1. Microcontroller Embedded C Programming: Absolute Beginners | Udemy
2. <https://a.impartus.com/ilc/#/course/1259590/1112>

Laboratory Session: Part A 6-12 Programs and Part B 1-5 Programs

Unit 5

Architecture of ARM Cortex-M, Cortex-M Processors, Cortex-M0, Modes and States, Programming Model, Memory Model, Nested Vector Interrupt Controller, Power management using sleep modes.

Pedagogy: Conceptual Learning using NPTEL video lectures for cortex systems

Recommended Resources / Links:

1. Microprocessors and Microcontrollers: Arm Cortex-M0+ (using RP2040) – NPTEL+
2. <https://a.impartus.com/ilc/#/course/1259590/1112>

Laboratory Session: Part B 6-9 Programs

Self-Learning:

Course-Project: Develop and Embedded system application using Raspberry Pi OR Arduino OR Nuvoton boards for to showcase real time working principle.

MICROCONTROLLER LAB

Part A

1. a. ALP to add first 10 odd numbers. Store sum in register.
b. ALP to compute sum of squares of 5 numbers starting from 1. Write and use procedure SQU. Store sum in register.
2. a. ALP to add the first n even numbers. Store the result in a memory location.
b. ALP to generate a geometric progression with a limit n. Display the results in memory.
3. a. ALP to count the number of zeroes and ones in a binary number.
b. ALP to find the average of ten 16-bit numbers stored in memory.
4. a. ALP to find the factorial of a number.
b. ALP to generate the first n Fibonacci numbers.
5. ALP to find the sum of digits of a number.
6. ALP to convert BCD number to binary.
7. ALP to select a set of r objects from a set of n objects without considering the order of elements in a selection using combination method.

8. ALP to select a set of r objects from a set of n objects considering the order of elements in an arrangement using permutation method.
9. ALP to implement Bubble Sort on an array of integers.
10. ALP to implement Binary Search on an array of integers.
11. ALP to check whether the given number is palindrome.
12. ALP to count the number of times a substring is repeated in the string.

Part B

1. C program to toggle the lowest pin of Port 0 with a delay between the two states. Observe and record the waveform obtained using the Logic Analyzer in the Keil simulator.
2. C program to generate an asymmetric square wave of 120Hz and having a duty cycle of 25% using the Timer0 module.
3. C program to generate a square wave using Timer0 in the interrupt mode.
4. C program to make a LED glow at different brightness levels (low to high) with brightness levels varying over duration of 2s. Demonstrate using logic analyzer window.
5. C program to display the string 'I LOVE ISE' in the serial window of UART1

Text Books:

1. Andrew Sloss, Dominic Symes, Chris Wright, "ARM System Developer's Guide – Designing and Optimizing System Software", Elsevier Publication 2012.
2. Lyla B Das, "Architecture, Programming, and Interfacing of Low-power Processors – ARM7, Cortex-M", Cengage Learning India Pvt. Ltd. 2017.

Reference Books:

1. F. Vahid and T. Givargis, "Embedded System Design: A Unified Hardware/Software Introduction", Wiley India Pvt. Ltd., 2002.
2. A.N. Sloss, D. Symes and C. Wright, "ARM System Developer's Guide: Design and Optimizing System Software", Morgan Kaufman Publishers, 2004.

Course Outcomes:

1. Understand the basics of embedded processors and operating system for different application scenarios (PO1, PSO2)
2. Describe the building blocks that contribute to the software aspects of embedded system design. (PO1, PO2, PO3, PSO1, PSO2)
3. Design ARM7 programs that includes interrupt structure and pipeline. (PO1, PO2, PO3, PO5, PSO1, PSO2)
4. Develop ARM7 assembly code for a given problem (PO1, PO2, PO3, PO5, PSO1, PSO2)

5. Compare Cortex-M processors in terms of architecture, memory model and interrupt structure
(PO1, PO2, PO3, PO5, PSO-2)

Course Assessment and Evaluation:

Continuous Internal Evaluation: 50 Marks		
Assessment Tool	Marks	Course Outcomes addressed
Internal test-I	30	CO1, CO2, CO3
Internal test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other components	Marks	Course Outcomes addressed
Course project	10	CO1, CO2, CO3, CO4, CO5
Lab test and Record	10	CO1, CO2, CO3, CO4, CO5
Semester End Examination	100	CO1, CO2, CO3, CO4, CO5

Design and Analysis of Algorithms	
Course Code: IS43	Credits: 3:0:0
Pre – requisites: NIL	Contact Hours: 45L+45S
Course Coordinator: Krishna Raj PM	

Course Contents

Unit 1

Foundations of Algorithms: Role of Algorithms in Computing, Growth of Functions, Asymptotic Notation, Analyzing Algorithms, Standard Notations and Common Functions. Divide-and-Conquer: Recurrences, Substitution Method, Recursion-Tree Method, Master Method. Probabilistic Analysis and Randomized Algorithms: Indicator Random Variables, Randomized QuickSort, Randomized Algorithms for Matrix Operations.

Pedagogy: Conceptual Learning using NPTEL video lectures and problem-solving sessions

Recommended Resources / Links: <https://nptel.ac.in/courses/106101060>

Unit 2

Sorting and Order Statistics: Heaps and HeapSort, Priority Queues, QuickSort, Sorting in Linear Time - Counting Sort, Radix Sort, Bucket Sort. Medians and Order Statistics: Selection Algorithms and their Analysis.

Pedagogy: Analytical Learning through NPTEL videos, illustrative case problems, and algorithm tracing

Recommended Resources / Links: <https://nptel.ac.in/courses/106101060>

Unit 3

Amortized Analysis and Advanced Algorithmic Techniques: Aggregate Analysis, Accounting Method, Potential Method. Union-Find / Disjoint Set Union with Path Compression and Union by Rank. Fibonacci Heaps and their applications in graph algorithms. Applications of Amortized Analysis to data structure operations and network optimization problems.

Pedagogy: Visualization-based Learning with animated NPTEL videos and hands-on paging examples

Recommended Resources / Links: <https://nptel.ac.in/courses/106101060>

Unit 4

Dynamic Programming: Matrix Chain Multiplication, Longest Common Subsequence, Optimal Binary Search Trees. Greedy Algorithms: Activity Selection, Huffman Codes, Minimum Spanning Trees using Kruskal and Prim, Dijkstra's Algorithm.

Pedagogy: Structure-oriented Learning using NPTEL lectures and whiteboard implementation exercises

Recommended Resources / Links: <https://nptel.ac.in/courses/106101060>

Unit 5

Graph Traversals: BFS, DFS, Topological Sort, Strongly Connected Components. Shortest Paths: Bellman-Ford, Dijkstra revisited, Floyd-Warshall, Johnson's Algorithm. Maximum Flow: Ford-Fulkerson, Edmonds-Karp. Computational Complexity: Introduction to P, NP, NP-Completeness, Approximation Algorithms.

Pedagogy: Real-world Case-based Learning with NPTEL content and practical scenario discussions
Recommended Resources / Links: <https://nptel.ac.in/courses/106101060>

Self-Learning:

Case Study: Analyze the use of a greedy algorithm in a real-world application like GPS navigation. Examine a scenario where an inefficient algorithm caused problems and propose a better solution.

Project Based Learning: Create a visualizer tool to demonstrate how sorting algorithms work. Build a simple recommendation engine using graph algorithms like BFS or DFS.

Simulation: Simulate Kruskal's Algorithm to find a minimum spanning tree in a network. Use a simulation to visualize the execution of a backtracking algorithm like the N-Queens Problem.

Text Books:

1. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, "Introduction to Algorithms," Fourth Edition, MIT Press, 2022.

Reference Books:

1. Jon Kleinberg, Éva Tardos, "Algorithm Design," 2nd Edition, Pearson, 2022.

2. Alfred V. Aho, John E. Hopcroft and Jeffrey D. Ullman, "Data Structures and Algorithms", Pearson Education, Reprint 2006.

Course Outcomes:

At the end of the course, the student will be able to:

1. Analyze algorithms using asymptotic notation, recurrences, and probabilistic tools. (PO1, PO2, PO4, PSO1)
2. Apply sorting, selection, and searching techniques efficiently. (PO1, PO3, PO5, PSO1)
3. Employ advanced data structures and amortized analysis to optimize algorithms. (PO1, PO3, PO5, PSO1)
4. Implement algorithmic design paradigms such as dynamic programming and greedy methods. (PO1, PO3, PO5, PSO1)
5. Apply graph algorithms, flow techniques, and complexity concepts to real-world computational problems. (PO2, PO4, PO11, PSO1)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Quiz	10	CO1, CO2, CO3, CO4, CO5
Simulation	10	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

DATABASE MANAGEMENT SYSTEMS	
Course Code: IS44	Credits: 3:0:0
Pre – requisites: Nil	Contact Hours: 45L+45S
Course Coordinator: Dr. Savita K Shetty	

Course Contents

Unit 1

Introduction to DBMS: Introduction, Characteristics of database approach, Actors on the scene, Workers behind the scene. Advantages of using DBMS approach. When not to use DBMS.

Database System Concepts: Data Models, Schemas and Instances, Three-schema architecture and data independence. Database languages and interfaces; Classification of Database Management systems.

Entity Relationship Model: Using High Level conceptual data for database design, an example database application. Entity Types, Entity Sets, Attributes and Keys; Relationship types, Relationship Sets, Roles and Structural Constraints; Weak Entity Types. Refining the ER Design; ER Diagrams, naming conventions and design issues.

Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation.

Recommended Resources / Links:

1. <https://www.youtube.com/watch?v=OWX4RvijwLw&list=PLIwC9bZ0rmjSkM1VRJROX4vP2YMIf4Ebh&index=4>
2. <https://www.youtube.com/watch?v=rrG7azSlyWI&list=PLIwC9bZ0rmjSkM1VRJROX4vP2YMIf4Ebh&index=4>

Unit 2

Relational Model and Relational Algebra: Relational Model Constraints and Relational Database Schemas; Update Operations, Transactions and dealing with constraint violations.

Unary relational Operations select and project. Relational algebra operations from set theory. Binary relational operations, join and division. Additional relational operations example of queries in relational algebra. Relational Database Design using ER- to-Relational Mapping.

Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation.

Recommended Resources / Links:

- <https://www.youtube.com/watch?v=DRSog3SA4Y&list=PLIwC9bZ0rmjSkM1VRJROX4vP2YMIf4Ebh&index=5>

Unit 3

SQL Data Definition and Data Types: The create table command in SQL, attribute data types and domains. Specifying basic attribute constraints and defaults in SQL; Specifying key and referential integrity constraints. Schema change statements in SQL; DROP command and Alter Command. Basic queries in SQL: The select from where structure of basic SQL queries, Unspecified where clause and use of the asterisk, Aliasing, Substring Pattern Matching and arithmetic Operators, Ordering of query results. Complex SQL Queries: Comparisons Involving null and three valued logics, nested queries, the exists and unique functions in SQL. Joined tables in SQL. Aggregate functions, Group by and having clauses. Insert, Delete and Update statements in SQL.

Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation.

Recommended Resources / Links:

1. <https://www.youtube.com/watch?v=w1XdPholzWY&list=PLIwC9bZ0rmjSkM1VRJROX4vP2YMIf4Ebh&index=6>
2. <https://www.youtube.com/watch?v=ywUn5qNmxNo&list=PLIwC9bZ0rmjSkM1VRJROX4vP2YMIf4Ebh&index=7>

Unit 4

Database Design: Informal Design Guidelines for Relation Schemas, reducing the null values in tuples, disallowing the possibility of generating the spurious tuples. Definition of Functional Dependencies, inference rules for functional dependencies. Normal Forms Based on Primary Keys-1NF, 2NF, 3NF, BCNF with examples. Codd rules.

Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation.

Recommended Resources / Links:

1. <https://www.youtube.com/watch?v=K5jqNjnE-pE&list=PLIwC9bZ0rmjSkM1VRJROX4vP2YMIf4Ebh&index=16>
2. <https://www.youtube.com/watch?v=9-uqoJK6KPc&list=PLIwC9bZ0rmjSkM1VRJROX4vP2YMIf4Ebh&index=17>

Unit 5

Introduction to Transaction Processing: Single user versus multi user transactions, Read and Write operations. Transaction and System Concepts, Desirable properties of Transactions, Transaction Support in SQL, Need of concurrency control and recovery. Two-phase Locking Techniques for Concurrency Control, Recovery techniques based on Deferred Update, Recovery techniques based on Immediate Update.

Introduction to NoSQL: Why NoSQL? characteristics and types of NoSQL database. Use of NoSQL in industry, SQL versus NOSQL.

Introduction to MongoDB: What is MongoDB? Why? Data Types and introduction to MongoDB Query Language.

Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation.

1. Recommended Resources / Links:

https://www.youtube.com/watch?v=TyT_MuPHG3g&list=PLIwC9bZ0rmjSkM1VRJROX4vP2YMIf4Ebh&index=31

2. <https://www.mongodb.com/resources/basics/databases/nosql-explained>

Text Books:

1. Ramez Elmasri and Shamkant B. Navathe: Fundamentals of Database Systems, 7th Edition, Addison-Wesley, 2016.
2. Benjamin Rosenzweig, Elena Silvestrova Rakhimov: Oracle PL/SQL by Example, 6th edition, 2023.

Reference Books:

1. Silberschatz, Korth and Sudharshan: "Data base System Concepts, Seventh Edition, Mc-GrawHill, 2019".
2. Raghu Ramakrishnan, Johannes Gehrke, "Database Management Systems, Third Edition", McGraw-Hill, 2003.
3. Arek Borucki "MongoDB 8.0 in Action 3rd Edition" 2025

Self-Learning (Case Study/ Project Based Learning): Students are required to work in teams of up to four members to complete a team-based project. The project must include both the development of a working database application and the submission of a project report for demonstration and evaluation purposes.

Each team should select a case study and follow the steps below to design and implement their application:

1. Collect Data Requirements
2. Gather Functional Requirements
3. Design the Entity-Relationship (ER) Model
4. Convert the ER Model to a Relational Schema and Normalize the Design

Course Outcomes:

At the end the course students will be able to:

1. Design Entity Relationship model for the given database problem. (PO1, PO2, PO3, PO9, PO10, PO11, PSO1, PSO-3)
2. Develop relational model from the given Entity Relationship diagram. (PO1, PO2, PO3, PO9, PO10, PO11, PSO1, PSO3)
3. Design the relational database schema by applying normalization techniques. (PO1, PO2, PO3, PO9, PO10, PO11, PSO1, PSO3)
4. Apply query language to perform database operations. (PO1, PO2, PO3, PO5, PO9, PO10, PO11, PSO1, PSO3)
5. Understand the concepts of transaction processing, concurrency control and recovery techniques. (PO1, PO11, PSO1, PSO3)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
Mini Project –Phase-1-Demo/Quiz	10	CO1, CO2, CO3,
Mini Project –Phase-2-Demo/Quiz	10	CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

ADVANCED JAVA	
Course Code: IS45	Credits: 3:0:0
Pre – requisites: Principles of Programming using C	Contact Hours: 45L+45S
Course Coordinator: Mrs Evangeline D	

Course Content

Unit I

Java Basics : Classes, Object, Constructors, Method overloading, Understanding Static, final, Inheritance-Super, Method Overriding, Dynamic Method Dispatch, Abstract Classes, Using Final. Packages and Interfaces- Defining, Implementing . Access Protection. Exception Handling- try-catch, throw, throws, finally. Built-in Exceptions, Creating Exception classes and Chained Exceptions

Pedagogy: Demo, Chalk and Talk.

Links:

- 1.<https://www.youtube.com/watch?v=VksxhzfD8kQ>
- 2.https://www.youtube.com/watch?v=4wZ_3oLoYiY

Unit II

String Handling: String Constructors, String Length, String Operations, changing case of characters, joining strings, string methods, Character Extraction, Searching Strings, Modifying Strings, Data conversions, StringBuffer. Type wrappers, Autoboxing

Lambda Expressions: Fundamentals, Functional Interfaces, block lambda expressions.

Multithreaded Programming : The Java Thread Model, The main thread, Creating a thread, Creating multiple threads, Using isAlive() and join()

Pedagogy: Demo, Chalk and Talk.

Links:

- 1.<https://www.youtube.com/watch?v=5NOLuoG6fcQ>
- 2.https://www.youtube.com/watch?v=IYuj9nn0PBU&list=PLfn3cNtmZdPOe3R_wO_h540QNfMkCQ0ho&index=30

Unit III

Collections: Collection Interfaces, Collection classes, Accessing a collection via an Iterator, Spliterator Storing user defined classes in Collections, Working with Maps, Sets, Comparators and Comparable Generics<T>: Introduction to Java Generics, Implementing Generics, Generic Methods and Interfaces

Pedagogy: Demo, Chalk and Talk.

Links: 1.https://www.youtube.com/watch?v=6rYOyIGfy3w&list=PLfn3cNtmZdPOe3R_wO_h540QNfMkCQ0ho&index=27

2. <https://www.youtube.com/watch?v=K1iu1kXkVoA>

Unit IV

Introducing Swing – The Swing packages, Event handling, Exploring Swing. **Working with Databases –** JDBC Drivers and Architecture, Accessing Databases,

Pedagogy: Demo, Chalk and Talk.

Links:

1. https://www.youtube.com/watch?v=ajhWv31oN1k&list=PLfn3cNtmZdPOe3R_wO_h540QNfMkCQ0ho&index=50
2. https://www.youtube.com/watch?v=mQj34vUhpts&list=PLfn3cNtmZdPOe3R_wO_h540QNfMkCQ0ho&index=43

Unit V

Java Servlets – Introduction, Servlet API and Life Cycle, JavaServer Pages Technology - What Is a JSP Page? The Life Cycle of a JSP Page, Creating Static Content, Creating Dynamic Content, Scripting in JSP Pages - Declarations, Scriptlets, Expressions

Pedagogy: Demo, Chalk and Talk.

Links:

1. <https://www.youtube.com/watch?v=7AljcZMo-V4>
2. <https://www.youtube.com/watch?v=OuBUUkQfBYM>

Text Books:

1. Herbert Schildt, Java:the Complete Reference, 12th edition, McGrawHill, , 2023.
2. Sharanam Shah, Vaishali Shah, JavaEE 7 for Beginners, Shroff Publishers, 2018.
3. Eric Armstrong, Jennifer Ball. Stephanie Bodoff, Debbie Bode Carson, Ian Evans, Dale Green, Kim Haase Eric Jendrock, J2EE 1.4 Tutorial, O'Reilly, 2005.

Reference Books:

1. Christian Ullenboom, Java: The Comprehensive Guide, Shroff Publishers, October, 2022

Self-Learning: Programming Test and GATE oriented aptitude test.

Course Outcomes:

At the end of the course, the student will be able to:

CO1: Apply basic Java concepts for various applications. (PO - 1, 2, 3, 4, 5 & PSO - 3)

CO2: Identify methods of Strings, type wrappers and Collection classes for applications (PO - 1, 2, 3, 4, 5 & PSO - 3)

CO3: Implement applications using Maps, Generics, Lambda and multithreading concepts (PO 1, 2, 3, 4, 5- & PSO - 3)

CO4: Develop programs using Swing and JDBC (PO 1, 2, 3, 4, 5 & PSO - 3)

CO5: Develop dynamic web pages using Servlets or JSP (PO -1, 2, 3, 4,5 & PSO - 3)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE) : 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
MCQ Test	10	CO1, CO2, CO3, CO4, CO5
Programming Test	10	CO1, CO2, CO3, CO4, CO5
Semester-End Examination (SEE)	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5

Design and Analysis of Algorithms Laboratory	
Course Code: ISL46	Credits: 0:0:1
Pre – requisites: NIL	Contact Hours: 30P
Course Coordinator: Krishna Raj	

Course Contents

General Instruction:

For each experiment, students are required to design and implement the algorithm, demonstrate the working through step-by-step tracing on sample input and evaluate its performance by analyzing the time complexity.

1. An online retail platform must arrange thousands of customer orders by order ID before shipping. Design an efficient sorting approach using arrays, ensuring stable ordering for large datasets.
2. A courier service wants to optimize package sorting for delivery by randomizing the partitioning step to avoid worst-case delays. Use array-based random partitioning to achieve efficient average performance.
3. An airline system maintains passenger boarding priorities based on ticket class and time of check-in. Implement a priority queue using a binary heap to order passengers for boarding efficiently.
4. An exam evaluation system needs to quickly sort thousands of student marks, all within the fixed range of 0 to 100. Apply a linear-time counting technique using arrays to produce the sorted list.
5. A hospital analytics dashboard requires finding the median patient waiting time from thousands of entries recorded daily. Use a selection procedure on an unsorted array to determine the median in guaranteed linear time.
6. A social networking platform must efficiently track friend groups as users form new connections. Implement a disjoint-set data structure with union by rank and path compression to dynamically manage merging groups.
7. A city navigation system computes the shortest routes between intersections for thousands of vehicles. Implement a graph representation with adjacency lists and use a heap-based priority queue optimized by Fibonacci heaps to improve shortest path calculations.
8. A video processing engine must determine the most efficient sequence for multiplying a chain of matrices representing video transformations. Use dynamic programming with a tabular approach to minimize the total number of scalar multiplications.
9. A plagiarism detection system compares two student essays to find the longest matching sequence of words. Use dynamic programming with a two-dimensional table to compute the longest common subsequence between the two text strings.
10. A telecom service provider compresses text messages to reduce transmission costs. Construct a binary tree based on character frequencies and apply a greedy encoding strategy to generate efficient prefix codes for the text.
11. A file management system must explore all files and folders in a directory structure to build dependency maps. Model the directory as a graph and traverse it using depth-first search to record discovery and finishing times.
12. A city water supply network distributes water from a main reservoir to multiple zones through pipelines of limited capacity. Model the network as a directed flow graph and compute the maximum possible water flow from source to sink.

Pedagogy: The reasoning-based, structured, and problem-oriented, aiming to build a deep conceptual understanding of algorithms rather than surface-level.

Links: https://www.youtube.com/playlist?list=PLyqSpQzTE6M9DKhN7z2fOpKTJWu-639_P

Text Books:

1. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, "Introduction to Algorithms," Fourth Edition, MIT Press, 2022.

Reference Books:

1. Jon Kleinberg, Éva Tardos, "Algorithm Design," 2nd Edition, Pearson, 2022.
2. Alfred V. Aho, John E. Hopcroft and Jeffrey D. Ullman, "Data Structures and Algorithms", Pearson Education, Reprint 2006.

Course Outcomes: At the end of the course, the student will be able to:

1. Design the experiment for the given problem using various standard algorithmic techniques. (PO1, PO2, PO3, PO9, PO10, PSO-1)
2. Develop the solution for the given real world problem. (PO1, PO2, PO3, PO4, PO9, PO10, PSO1)
3. Analyze the results and produce substantial written documentation (PO1, PO2, PO4, PO9, PO10, PSO1)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test	20	CO1, CO2, CO3
Other Components		
Conduction, Calculation, Record and Viva	30	CO1, CO2, CO3
Semester-End Examination (SEE)	50	CO1, CO2, CO3

DATABASE MANAGEMENT SYSTEMS LAB	
Course Code: ISL47	Credits: 0:0:1
Pre – requisites: Nil	Contact Hours:30P
Course Coordinator: Dr. Savita K. Shetty	

List of Experiments:

Part A

1. Consider an Employee with a social security number (SSN) working on multiple projects with definite hours for each. Each Employee belongs to a Department. Each project is associated with some domain areas such as Database, Cloud and so on. Each Employee will be assigned to some project. Assume the attributes for Employee and Project relations.
 - a) Mention the constraints neatly.
 - b) Design the ER diagram for the problem statement
 - c) State the schema diagram for the ER diagram.
 - d) Create the tables, insert suitable tuples and perform the following operations in SQL i. Obtain the details of employees assigned to “Database” project.
 - ii. Find the number of employees working in each department with department details.
 - iii. Update the Project details of Employee bearing SSN = #SSN to ProjectNo = #Project_No and display the same.

2. Consider the relations: PART, SUPPLIER and SUPPLY. The Supplier relation holds information about suppliers. The attributes SID, SNAME, SADDR describes the supplier. The Part relation holds the attributes such as PID, PNAME and PCOLOR. The Shipment relation holds information about shipments that include SID and PID attributes identifying the supplier of the shipment and the part shipped, respectively. The Shipment relation should contain information on the number of parts shipped.
 - a) Mention the constraints neatly.
 - b) Design the ER diagram for the problem statement
 - c) State the schema diagram for the ER diagram.
 - d) Create the above tables, insert suitable tuples and perform the following operations in Oracle SQL:
 - i. Obtain the details of parts supplied by supplier #SNAME.
 - ii. Obtain the Names of suppliers who supply #PNAME.
 - iii. Delete the parts which are in #PCOLOR.

3. Consider the relations BOAT, SAILOR and RESERVES. The relation BOAT identifies the features of a boat such as unique identifier, color and a name. The list of sailors with attributes such as SailorID, name, age etc., are stored in the relation SAILOR. The sailors are allowed to reserve any number of boats on any day of the week and the records are to be updated in the RESERVES table.
 - a) Mention the constraints neatly.
 - b) Design the ER diagram for the problem statement
 - c) State the schema diagram for the ER diagram.
 - d) Create the tables, insert suitable tuples and perform the following operations in SQL:
 - i. Obtain the details of the boats reserved by ‘#Sailor_Name’.
 - ii. Retrieve the BID of the boats reserved necessarily by all the sailors.
 - iii. Find the number of boats reserved by each sailor. Display the Sailor_Name along with the number of boats reserved.

4. Consider the Banking database – CUSTOMER, BRANCH, ACCOUNT and TRANSACTION. An account can be a savings account or a current account. Customer can have both types of accounts. The transactions can be a deposit or a withdrawal.
 - a) Mention the constraints neatly.
 - b) Design the ER diagram for the problem statement
 - c) State the schema diagram for the ER diagram.
 - d) Create the above tables, insert suitable tuples and perform the following operations in SQL:
 - i. Obtain the details of customers who have both Savings and Current Account.
 - ii. Retrieve the details of branches and the number of accounts in each branch.

- iii. Obtain the details of customers who have performed at least 3 transactions.
 - iv. List the details of branches where the number of accounts is less than the average number of accounts in all branches.

- 5 Consider the Book Lending system from the library- BOOKS, STUDENT, BORROWS. The students are allowed to borrow any number of books on a given date from the library. The details of the book should include ISBN, Title of the Book, author, and publisher. All students need not compulsorily borrow books.
 - a) Mention the constraints neatly.
 - b) Design the ER diagram for the problem statement
 - c) State the schema diagram for the ER diagram.
 - d) Create the above tables, insert suitable tuples and perform the following operations in SQL:
 - i. Obtain the names of the student who has borrowed either book bearing ISBN '123' or ISBN '124'.
 - ii. Obtain the Names of female students who have borrowed "Database" books.
 - iii. Find the number of books borrowed by each student. Display the student details along with the number of books.

- 6 Consider the Hospital management system with Patient, Doctor and Appointment. Patient relation holds the Patient ID, Name, Age, Gender. Doctor relation describes with Doctor ID, Name, Specialization Each patient has given appointment with some doctor on specific date.
 - a) Mention the constraints neatly.
 - b) Design the ER diagram for the problem statement
 - c) State the schema diagram for the ER diagram.
 - d) Create the above tables, insert suitable tuples and perform the following operations in Oracle SQL:
 - i. Retrieve the details of doctors consulted by '#Patient_Name'.
 - ii. Find the DoctorIDs of doctors consulted by **all patients**.
 - iii. For each patient, find the number of doctors consulted. Display Patient_Name with the count.

- 7 a) Create an Employee and Department tables, insert suitable tuples and perform the following operations using MongoDB
 - i. List all the employees of Department named #Dept_name
 - ii. Name the employees working on Project Number :#Project_No
 b) Write a program that gives all employees in Department #number a 15% pay increase. Display a message displaying how many employees were awarded the increase.

- 8 a) Create Patient and Doctor collections, insert suitable documents, and perform the following operations using MongoDB:
 - i. List all the patients treated by the doctor named #Doctor_Name.
 - ii. Display the names of patients who have an appointment on date #Appointment_Date.
 b) Write a program that increases the consultation fee of all doctors in specialization #Specialization by 20%. Display a message showing how many doctors were updated.

- 9 a) Create the table, insert suitable tuples and perform the following operations using MongoDB
 - i. Update the details of parts for a given part identifier: #PID.
 - ii. Display all suppliers who supply the part with part identifier: #PID.
 b) Write a PL/SQL program to copy the contents of the Shipment table to another table for maintaining records for specific part number.

- 10 a) Create the table, insert suitable tuples and perform the following operations using MongoDB.
 - i. Obtain the number of boats obtained by sailor :#Sailor_Name
 - ii. Retrieve boats of color :'"#color"'
 b) Consider the following relation schema.
 EMPLOYEE (SSN, Name, sal, DeptNo)
 Write a trigger to raise an error if the table is modified on a specific day (Eg., Saturday or Sunday) of the week.

- 11 a) Create the tables Customer and Branch, insert suitable tuples and perform the following operations using MongoDB
 - i. Find the branch name for a given Branch_ID.
 - ii. List the total number of accounts for each customer.
- b) Using cursors demonstrates the process of copying the contents of one table to a new table.
- 12 a) Create the table, insert suitable tuples and perform the following operations using MongoDB
 - i. Obtain the book details authored by “author_name”.
 - ii. Obtain the Names of students who have borrowed “Database” books.
- b) Create a program that accepts two numbers. If the first is larger than the second raise an exception called e_bigger and display an appropriate message.

Part B

Mini Project

Students are required to work in teams of up to four members to complete a team-based project. The project must include both the development of a working database application and the submission of a project report for demonstration and evaluation purposes.

Each team should select a case study and follow the steps below to design and implement their application:

5. Use SQL for Data Definition:
 - o Create tables based on the relational schema
 - o Populate the tables with sample data
6. Use SQL for Data Manipulation
7. Create Views and Triggers using SQL
8. Execute Basic SQL Queries
9. Write and Execute Complex SQL Queries
10. Design and Implement a Front-End Interface for the application

Text Books:

1. Ramez Elmasri and Shamkant B. Navathe: Fundamentals of Database Systems, 7th Edition, Addison-Wesley, 2016.
2. Benjamin Rosenzweig, Elena Silvestrova Rakhimov: Oracle PL/SQL by Example, 6th edition, 2023.

Reference Books:

1. SQL/PLSQL –The programming Language of Oracle by Evan Byrass 4TH Edition BPB publication 2009

Course Outcomes:

At the end of the course the student will be able to:

1. Develop relational database schema by using data definition language utilities. (PO1, PO2, PO3, PO4, PO5, PO9, PO10, P11, PSO1, PSO3)
2. Develop solutions to the broad range of query and data update problems using SQL/MongoDB, (PO1, PO2, PO3, PO4, PO5, PO9, PO10, P11, PSO1, PSO3)
3. Implement programmatic components for a database using PLSQL (PO1, PO2, PO3, PO4, PO5, PO9, PO10, PO11, PSO1, PSO3)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test	20	CO1, CO2, CO3
Other Components		
Conduction, Project Work, Record and Viva	30	CO1, CO2, CO3
Semester-End Examination (SEE)	50	CO1, CO2, CO3

ADVANCED JAVA LAB	
Course Code: IS48	Credits: 0:0:1
Pre – requisites: Nil	Contact Hours: 30P
Course Coordinator: Mrs. Evangeline D	

Course Content

List of Experiments

- Write a Java Program for the following Payroll System. Create a package called PayrollManagement. Define an interface Payable with a method calculate(). Create a super class called Employee with the following data members name, age, salary and designation. Use a parameterised constructor to initialize all the data members. The Employee class is inherited in three sub-classes namely FullTime, PartTime and Intern and these subclasses implement the Payable interface. Full-Time Employee receives monthly salary and Part-Time Employee receives hourly wages. Intern takes a fixed stipend. Create another package “Company” and import these classes to complete the payroll system. Create an array of Employee objects.
- Create a Queue class and implement enqueue, dequeue and display operations. Create Custom Exceptions to handle Queue Overflow and Queue Underflow.
- Define a package named stringoperations to encapsulate the string processing functionality. Create an interface named StringManipulator with at least eight methods for string operations:
 - String reverse(String input): Reverses the given string.
 - String toUpperCase(String input): Converts the string to uppercase.
 - String toLowerCase(String input): Converts the string to lowercase.
 - String concatenate(String str1, String str2): Concatenates two strings.
 - int countVowels(String input): Counts the number of vowels in the string.
 - int wordCount(String input): Counts the number of words in the string.
 Implement the interface in a class named StringProcessor within the stringoperations package, providing concrete implementations for all methods.
 Create a main class in a separate package application, where:
 - An object of StringProcessor is created.
 - The user is prompted to enter a string.
 - Each method of the interface is tested and its output is displayed.
- Create a Palindrome Checker program using StringBuffer that verifies if a given string is a palindrome. The program should implement user-defined exception handling for the following:
 If the input string contains non-alphabetical characters, throw a custom exception InvalidInputException.
 If the string length is less than 3 characters, throw a custom exception ShortStringException.
 If the string is a palindrome (irrespective of the case), print a message indicating that it is a palindrome.
 If the string is not a palindrome, print a message indicating that it is not a palindrome.
- Create a Password Security Application in Java that takes a user's password as input and performs the following operations .
 - Check if the password contains at least one uppercase letter, one lowercase letter, and one digit.
 - Count the number of special characters in the password.
 - Mask the password by replacing all characters with * (except the first and last character).
 - Reverse the password and display it (for security encryption demonstration).
 - Append a random security token (e.g., "@123!") to the password.
 - Replace all vowels (a, e, i, o, u) with # to make it difficult to read.

6. Write a Program that simulates a telephone that records missed incoming calls. For each missed call, store the time of call, telephone number of origin, and name of the caller if the name is available. For unlisted numbers, set the name to “private caller”. Choose or extend the most appropriate collection class and provide the following features.
 - a. Numbers are recalled in the order they arrive
 - b. Up to 3 numbers are recorded. When the fourth call comes in, it is stored and the oldest call is deleted so that no more than 3 numbers are ever recorded.
 - c. After each number display, the user can
 - i. Identify whether the number has to be deleted or not
 - ii. To display the call details (number, caller name and time).

Delete the list of numbers specified by the user as in i.

Write a helper class to represent an incoming call with fields to hold the number, name of the caller, and time of the call. Write a tester call that stores the several numbers, simulate the user pressing the missed-calls button, and finally prints the entire collection of stored calls.
7. Write a Java program using user-defined storage classes to create a book database and store it in a Collection List.
 - a. Books collection should include title, author, publisher and price.
 - b. Write a method to sort the books in ascending order of price and store it in another List. Maintain the book details with respect to a unique book id.
 - c. Prompt for an author name and list all the books with the same author name. Create a new list holding all the book details with price greater than a user specified price.
8. Write a program to create generic Stack class with push(), pop(), clear(), isEmpty() and display() methods. Demonstrate creating Stack of String and Integer objects.
9. Create a desktop java application using swings to enable a user to enter student information such as name, usn, age, address, sgpa of 4 semesters, category.
 - a. Perform validations on all the fields. Display appropriate messages in pop up boxes to indicate wrong entries.
 - b. On clicking of the “compute” button, find the cgpa (Average of sgpas) . On clicking of the “done” button, place the student details (name, usn, age, address,cgpa, category) in a collection.
 - c. Display the collection in a textarea on the click of a “complete” button. Enable / Disable the visibility of “compute” and “done” buttons.
10. Write a java program using Swing to validate user login information using dialog boxes.
 - a. Once validated, allow the user to enter the customer id, if the person is a new customer, else check whether the customer exists in a collection and obtain the customer id.
 - b. The customer id can be obtained given a mobile number. Allow the user to enter the item purchased by giving the item id and quantity purchased.
 - c. On clicking of a button, the item name and the total cost should appear in the corresponding GUI components.
 - d. Using option dialog box, indicate the types of discount available for the customer. On clicking on the print button, print the details in information dialog box.
11. Write a program that uses Java Swing and JDBC to create a stand-alone application:
 - a. Create two tables namely, Representative (RepNo, RepName, State, Comission, Rate) and Customer (CustNo, CustName, State, Credit_Limit, RepNo) in MySQL database. Use appropriate Swing components to insert values in a form.
 - b. Use another form to display Representative’s information when Credit_Limit is above 15,000.
12. Create a JSP based web application for purchasing a type of shirt from a list of shirts. Display a table giving the type of shirts and cost. With a drop down menu, select the type of shirt. Use radio-buttons to select the neck type of shirt (Round neck, V neck and collar).

In a textbox, enter the quantity. Include “compute” button that calculates the total cost of purchase done and displays the purchase details. Include appropriate labels.

Text Books:

1. Herbert Schildt, Java:the Complete Reference, 12th edition, McGrawHill, , 2023.
2. Sharanam Shah, Vaishali Shah, JavaEE 7 for Beginners, Shroff Publishers, 2018.
3. Eric Armstrong, Jennifer Ball. Stephanie Bodoff, Debbie Bode Carson, Ian Evans, Dale Green, Kim Haase
Eric Jendrock, J2EE 1.4 Tutorial, O’Reilly, 2005.

Reference Books:

1. Christian Ullenboom, Java: The Comprehensive Guide, Shroff Publishers, October, 2022

Course Outcomes:

At the end of the course the student will be able to:

CO1: Apply Collection Framework Generics & Swings to solve given problem (PO - 1, 2, 3, 4, 5, & PSO - 3)

CO2: Apply JDBC concepts to work with structured data (PO - 1, 2, 3, 4, 5 & PSO - 3)

CO3: Apply JSP and Servlet to develop web applications (PO - 1, 2, 3, 4, 5 & PSO - 3)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test	20	CO1, CO2, CO3
Other Components		
Conduction, Execution, Record and Viva	30	CO1, CO2, CO3
Semester-End Examination (SEE)	50	CO1, CO2, CO3

UI/UX	
Course Code: ISAE491	Credits: 1:0:0
Pre – requisites: Nil	Contact Hours: 15L+15S
Course Coordinator: Dr. Anitha K	

Course Content

Unit I

The importance of the user interface: Defining the user interface, the importance of good design, the benefits of good design. **Characteristics of graphical and web user interfaces:** interaction styles, the graphical user interface, the concept of direct manipulation, graphical system : advantages and disadvantages, characteristics of the graphical user interface, the web interface, the merging of graphical business system, principles of user interface design

- Pedagogy/Course Delivery tools: Chalk and talk, Power Point Presentation
- Link:1. https://www.youtube.com/watch?v=cYVkbo_OIFQ&t=461s
- 2. https://www.youtube.com/watch?v=c9Wg6Cb_YIU

Unit II

The user interface design process : usability, **Know your user or client:** important human characteristic in design, **Understand the principles of good interface and screen design:** navigation flow, Visually pleasing composition, application and page size, application screen elements.

- Pedagogy/Course Delivery tools: Chalk and talk, Power Point Presentation
- Link: https://www.youtube.com/watch?v=_6TI2_eM0DE

Unit III

Develop system menus and navigation schemes: Structures of Menus, Functions of Menus, Content of Menus, Phrasing the Menu, Formatting of Menus, Selecting Menu Choices, Web Site Navigation, and Kinds of Graphical Menus **select the proper kinds of windows:** Window Characteristics, components of windows, Window Presentation Styles, Types of Windows, Organizing Window Functions, The Web and the Browser.

- Pedagogy/Course Delivery tools: Chalk and talk, Power Point Presentation
- Links: <https://www.youtube.com/watch?v=jmexGpXLtVc>

Unit IV

Interactive devices: Input Devices, output devices. **Choose the proper screen – based controls:** Operable Controls, Text Entry/Read-Only Controls, Selection Controls, Combination Entry/Selection Controls, Custom Controls, Presentation Controls, Selecting the Proper Controls.

- Pedagogy/Course Delivery tools: Chalk and talk, Power Point Presentation
- Links:1. <https://www.youtube.com/watch?v=E-Lzclhyx4c>
- 2. <https://www.youtube.com/watch?v=uABgEB0EILI>

Unit V

Organize and layout windows and pages: Organizing and Laying Out Screens. **Test, test and retest:** Usability, Prototypes, Kinds of Tests, Developing and Conducting a Test.

- Pedagogy/Course Delivery tools: Chalk and talk, Power Point Presentation.
- Links: 1. <https://www.youtube.com/watch?v=ZcJsd-3Yi7g>
- 3. <https://www.youtube.com/watch?v=OlbdIXLunt4>

Self Learning- Practical Based learning

- Design UI for web or mobile app.

Suggested Learning Resources:**Text Books:**

1.The essential guide to user interface design, third edition, Wilbert O Galitz, 2007

Reference Book:

1.Rex Hartson, Pardha S Pyla, The UX Book, Process and Guidelines for Ensuring a Quality User Experience, Morgan Kauffman, 2012

Course Outcomes (COs):

At the end of the course, the student will be able to:

CO1: understand the importance of User Experience and User Interface design in application development (PO - 2, 3, 5, 9, 10 & PSO - 1)

CO2: Select and use appropriate interactive devices and screen-based controls in the design of a user interface, based on the needs and goals of the user. (PO - 2, 3, 5, 9, 10 & PSO - 1)

CO3: Design system menus and navigation schemes that are intuitive and user-friendly. (PO - 2, 3, 5, 9, 10 & PSO - 1)

CO4: Select appropriate types of windows, interactive devices, and screen-based controls to enhance user interaction and improve overall user experience. (PO - 2, 3, 5, 9, 10 & PSO - 1)

CO5: Applying feedback to iterative testing processes, refine and improve the UI design continuously. (PO - 2, 3, 5, 9, 10 & PSO - 1)

Course Assessment and Evaluation:

Continuous Internal Evaluation: 50 Marks		
Assessment Tool	Marks	Course outcomes addressed
Internal test-I	30	CO1, CO2, CO3
Internal test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other components	Marks	Course outcomes addressed
Practical Based learning	20	CO1,CO2,CO3, CO4, CO5

Data Analytics using R Programming	
Course Code: ISAE492	Credits: 1:0:0
Pre – requisites: NIL	Contact Hours:15L+15S
Course Coordinator: Dr. Pushpalatha M N	

Course Contents

Unit 1

Introduction to R- Introduction, Downloading and Installing R, IDEs and Text Editors, Handling Packages in R, Getting Started with R- Introduction, Working with Directory, Data Types in R, Few Commands for Data Exploration.

Pedagogy: Demo, Chalk and Talk

Links: 1. <https://www.youtube.com/watch?v=3iSKFCKLUsl> 2. <https://www.youtube.com/watch?v=GM4D2Z-5TU4>
3. <https://www.youtube.com/watch?v=pJj1T35kaGo>

Unit 2

Loading and Handling Data in R- Introduction, Challenges of Analytical Data Processing, Expression, Variables and Functions, Missing Values Treatment in R, Using the ‘as’ Operator to Change the Structure of Data, Vectors, Matrices, Factors, List, Few Common Analytical Tasks.

Pedagogy: Demo, Chalk and Talk

Links: 1. <http://www.digimat.in/nptel/courses/video/106106179/L09.html>
2. <https://www.youtube.com/watch?v=vwMltOwTPBA>

Unit 3

Aggregating and Group Processing of a Variable, Simple Analysis Using R, Methods for Reading Data, Comparison of R GUIs for Data Input, Using R with Databases and Business Intelligence Systems. Pedagogy: Demo, Chalk and Talk

Links: <https://www.youtube.com/playlist?list=PLelwGZPgtiz5bZmBWNN1GwVo2Ly1SdWEg>

Unit 4

Exploring Data in R- Introduction, Data Frames, R Functions for Understanding Data in Data Frames, Load Data Frames, Exploring Data, Data Summary, Finding the Missing Values, Invalid Values and Outliers, Descriptive Statistics.

Pedagogy: Demo, Chalk and Talk

Links: 1. <https://www.youtube.com/watch?v=t4bTmlZLfaw>

Unit 5

Spotting problems using graphics and visualization- Visually checking distributions for a single variable, visually checking relationships between two variables, Data visualization in R Basic graphics- Scatter, Bar and Line. **Managing data:** Cleaning data- Treating missing values (NAs), Data transformations, Sampling for modeling and validation

Pedagogy: Demo, Chalk and Talk

Link: <https://www.youtube.com/watch?v=dJyzJRFQj90>

Text Books:

1. Data analytics using R, Seema Acharya, McGraw Hill Education (India) Private Limited.2018.
2. Practical Data Science with R, Nina Zumel and John Mount, Manning Publication. 2014.

Reference Books:

1. R for Data Science: Import, Tidy, Transform, Visualize, and Model Data 1st Edition, Hadley Wickham and Garrett Grolemund, 2017

Self-Learning- Programming Assignment**Course Outcomes:**

CO1: Understand and apply basic concepts of R programming, including variables, data types, and data frames.

CO2: Apply logical and matrix operations, and develop programs in R using functions and control structures.

CO3: Load, explore, and summarize data from files or URLs in R to identify issues such as missing values, invalid entries, and outliers.

CO4: Use graphical techniques in R to visualize data distributions, explore relationships between variables, and identify potential data issues using basic plots like scatter, bar, and line charts.

CO5: Manage and clean data in R by handling missing values, performing transformations, and applying sampling techniques for modeling and validation.

Course Assessment and Evaluation:

Continuous Internal Evaluation: 50 Marks		
Assessment Tool	Marks	Course outcomes addressed
Internal test-I	30	CO1, CO2, CO3
Internal test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other components	Marks	Course outcomes addressed
Programming Assignment	20	CO1,CO2,CO3, CO4, CO5

**IV Semester
(For Lateral Entry Diploma Students)**

ADDITIONAL MATHEMATICS – II

Course Code: AM41

Credits: 0:0:0

Contact Hours: 42

Course coordinators: Dr. Veena B N

Pre-requisites: Nil

Unit I

Differential Calculus- I: Partial differentiation, Euler's theorem, total differential coefficient, differentiation of composite and implicit functions.

- Pedagogy / Course delivery tools: ➤ Chalk and talk
- Links: ➤ <https://nptel.ac.in/courses/111/105/111105121/>
- Impartus recording: ➤ <https://a.impartus.com/ilc/#/course/107625/1030>

Unit II

Differential Calculus- II: Jacobian and Properties. Taylor's theorem for function of two variables, maxima and minima for functions of two variables.

- Pedagogy / Course delivery tools: ➤ Chalk and talk
- Links: ➤ <https://nptel.ac.in/courses/111/105/111105121/>
- Impartus recording: ➤ <https://a.impartus.com/ilc/#/course/107625/1030>
➤ <https://a.impartus.com/ilc/#/course/59742/295>

Unit III

Vector Integration: Line integrals, surface integrals and volume integrals. Green's theorem, Stokes' and Gauss divergence theorem (without proof) and problems, orthogonal curvilinear coordinates.

- Pedagogy / Course delivery tools: ➤ Chalk and talk
- Links: ➤ <https://nptel.ac.in/courses/111/105/111105134/>
- Impartus recording: ➤ <https://a.impartus.com/ilc/#/course/107625/1030>

Unit IV

Higher Order Differential Equations: Higher order linear differential equations, method of variation of parameters, Cauchy's and Legendre's homogeneous differential equations.

- Pedagogy / Course delivery tools: ➤ Chalk and talk
- Links: ➤ <https://nptel.ac.in/courses/111/105/111105121/>
- Impartus recording: ➤ <https://a.impartus.com/ilc/#/course/96127/452>
➤ <https://a.impartus.com/ilc/#/course/59742/295>

Unit V

Probability: Introduction. Sample space and events. Axioms of probability. Addition and multiplication theorems. Conditional probability- illustrative examples. Bayes theorem – examples.

- Pedagogy / Course delivery tools: ➤ Chalk and talk
- Links: ➤ <https://nptel.ac.in/courses/111/107/111107119/>
- Impartus recording: ➤ <https://a.impartus.com/ilc/#/course/283623/703>

Self Learning:

Problem solving and Quiz.

Text Books:

1. **B.S. Grewal** – Higher Engineering Mathematics, Khanna Publishers, 44th edition, 2017.
2. **Erwin Kreyszig** – Advanced Engineering Mathematics – Wiley Publication, 10th Edition, 2015.

Reference Books:

1. **H. K. Dass** – Higher Engineering Mathematics – S Chand Publications, 1998
2. **B. V. Ramana** – Engineering Mathematics – Tata McGraw-Hill Publishing Co. Ltd., New Delhi, 2008.

Course Outcomes

At the end of the course the student will be able to

1. To carryout differentiation of function of several variables. (PO-1,2)
2. Solve the problems related to Jacobians, the extreme values of a function and Taylors series. (PO-1,2)
3. Exhibit the interdependence of line, surface and volume integrals using integral theorems. (PO-1,2)
4. Find the solution of second and higher order ODEs with constant and variable coefficients. (PO-1,2)
5. Solve the problems on conditional probability and Baye's theorem. (PO-1,2)

Course Assessment and Evaluation:

Continuous Internal Evaluation: 50 Marks		
Assessment Tool	Marks	Course outcomes addressed
Internal test-I	30	CO1, CO2, CO3
Internal test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other components	Marks	Course outcomes addressed
Quiz	10	CO1, CO2, CO3
Assignment	10	CO3, CO4, CO5